When Do Foundation Models Pay Off for Retail Demand Forecasting?
A Benchmark Study with Structured Literature Search

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Abstract

We study when zero-shot time-series foundation models improve short-term retail demand forecasts relative to conventional baselines. The study combines a structured extraction from fev-bench and GIFT-Eval with matched-panel local experiments on M5, Rohlik, and Favorita. In the external benchmark extraction ($k = 498$ FM-vs-best-baseline contrasts), SQL shows the clearest directional FM advantage (median log-ratio -0.229; FM lower in 92% of rows), whereas WAPE/ND (median -0.047; 66%) and MASE (median -0.036; 61%) are much closer to conventional baselines. Because the extracted series-count term is a workload proxy rather than a validated sampling variance, these cross-suite results are descriptive rather than a confirmatory pooled meta-analysis. For prediction-level evidence, we rerun official fev-bench retail windows and compute paired bootstrap WAPE covariance matrices for 20 retail tasks. In Source B, we evaluate five FMs and six conventional baselines on shared 100-series rolling-origin panels with paired bootstrap log-ratio covariance. Chronos-2 is below the best observed local baseline in all nine M5, Rohlik, and Favorita dataset–horizon cells; Moirai 2.0-Small and TiRex are lower in all nine point estimates and have intervals excluding zero in seven. A harder M5 panel stratified by volume and zero fraction preserves the Chronos-2 advantage over the best local LGBM baseline at $h = 7, 14, 28$. The practical conclusion is conditional: rankings depend on metric, baseline strength, multi-step protocol, workload matching, leakage risk, and demand characteristics. We release prediction-level artefacts, covariance matrices, and a validation checklist for retail bake-offs.

Keywords: demand forecasting, foundation models, benchmark study, LightGBM, structured literature search, retail, time series

1. Introduction

The 2024–2026 wave of time-series foundation models — Ansari et al. (2025), TimesFM 2.5 (Google Research, 2025), Moirai 2.0 (Liu et al., 2025), TiRex (Auer et al., 2025) — has shifted the default assumption in short-term forecasting. Where a retail team in 2022 would start with LightGBM and a feature-engineering sprint, the 2026 playbook increasingly says: try a zero-shot foundation model first, tune LightGBM only if the FM disappoints. Each model

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paper reports favourable results on its own evaluation suite — GIFT-Eval, fev-bench, or a curated subset of the M competitions — but no paper runs the same battery of models under matched conditions on the same retail datasets and reports the delta. The benchmarks multiply; the synthesis does not.

This paper asks: **when do foundation models actually pay off for retail demand forecasting?** We operationalise "pay off" as an observed, reproducible reduction in forecast error relative to a clearly specified conventional baseline (gradient-boosted tree or statistical model), evaluated under conditions that are as comparable as the available benchmark evidence allows. By "when" we mean: on which datasets, horizons, demand distributions, metrics, and hardware budgets.

1.1. Research questions

We address four research questions across three complementary lenses (accuracy, moderators, and cost):

- **RQ1 (Accuracy):** Do zero-shot foundation models outperform conventional baselines on retail demand forecasting? (§6.2)
- **RQ2 (Moderators):** Which characteristics — metric choice, demand intermittency, model scale, forecast horizon — moderate the FM-vs-baseline gap? (§6.8)
- **RQ3 (Cost):** What cost-error tradeoffs appear when consumer-hardware electricity, cloud-GPU list price, and CO_2 footprint are included? (§6.7)
- **RQ4 (Guidance):** Under what conditions should a retail forecasting team deploy a foundation model instead of investing in feature engineering for LightGBM? (§8)

1.2. Approach

This study is a benchmark study supported by a structured literature search. The search defines the retail evidence gap and documents which published FM evaluations are extractable; the main evidence comes from benchmark extraction and matched-panel experiments. Because the paired statistical mass comes mainly from fev-bench and GIFT-Eval, the paper reports metric-specific, suite-specific, and leave-one-suite-out summaries rather than a single confirmatory pooled claim.

1. **Source A (literature + benchmark extraction).** We use 150 identified records (2020–2026) from retained Google Scholar/web, Semantic Scholar, and arXiv searches, applying retail-domain inclusion criteria. Scopus and Web of Science target strings are reported as non-retained search limitations rather than exportable logs. We supplement the extraction with per-model per-task results from fev-bench (Shchur et al., 2025) (20 retail tasks, 7 FMs vs LightGBM/CatBoost baselines) and GIFT-Eval (Aksu et al., 2024) (4 Sales-domain tasks, 7 FMs vs statistical baselines). The frozen extraction schema contains 802 rows across 9 retail datasets (§3.2).

2. **Source B (gap-filling experiments).** We run our own experiments under a unified evaluation protocol (§5.1): five foundation models (Chronos-Bolt-Tiny, Moirai 2.0-Small, TiRex, Chronos-2, TimesFM 2.5; 9–200 M parameters) on a consumer MacBook (Apple Silicon MPS / CPU), and three baselines (lightgbm_cov, lightgbm_direct, seasonal_naive) on Azure ML. Source B produces matched-panel rows with runtime, cost, and CO_2 metadata for a 100-series M5, Rohlik, and Favorita panel. The final analysis reports this shared panel for Chronos-Bolt-Tiny, Chronos-2, Moirai 2.0-Small, batched TiRex, TimesFM 2.5, and the baseline variants. Earlier unmatched Source B sweeps are retained only as provenance and data-integrity material in the appendices, not as primary evidence.

The primary benchmark analysis (§6) uses Source A benchmark rows: for each (FM, task, metric) pair we compute log(FM metric/best conventional baseline metric). The primary file has $k = 498$ Source A pairs. We also generate heuristic weighted `rma.mv` diagnostics for continuity, but the $1/n_{\mathrm{FM}} + 1/n_{\mathrm{baseline}}$ term is treated as a workload proxy rather than as a validated sampling variance. The matched-panel Source B experiments are reported separately because they have their own prediction-level paired bootstrap covariance matrices.

1.3. Headline findings

Three findings motivate the sections that follow:

1. **The observed FM advantage is benchmark-level and metric-dependent.** Across $k = 498$ Source A benchmark pairs, SQL has the clearest directional advantage (median log-ratio -0.229; FMs lower in 92% of rows), while WAPE/ND and MASE are closer to the best baseline (median log-ratios -0.047 and -0.036). The finding is therefore an observed benchmark-corpus pattern, not a universal claim.

2. **Model scale is not a stable predictor in the current benchmark corpus.** Source B point estimates are clustered across several small and medium FMs, while Source A does not yield a stable scale moderator. Because Source B is not fully paired, this is a hypothesis about the absence of a simple size–accuracy gradient, not evidence that one named checkpoint dominates another. This is consistent with Shchur et al. (2025)'s finding that architecture may matter more than parameter count.

3. **The LightGBM protocol gap varies across panels, metrics, and budgets.** §5.3 shows that the direct-vs-recursive multi-step gap changes with dataset, workload, metric aggregation, and compute budget. The matched aggregate-WAPE panel makes M5 close at short horizons and mixed on Favorita, while per-series WAPE is much more sensitive to small denominators. Intermittency is a plausible stratification variable, but the evidence supports a protocol hypothesis to validate, not a hard threshold rule. No retail-benchmark protocol recommendation that ignores demand distribution, metric choice, and compute budget should be treated as general.

1.4. Contributions

1. **Structured literature search** of foundation models for retail demand forecasting, used to document the extractability gap and motivate the benchmark study (802 extraction-schema rows, 9 datasets, 2020–2026).

2. **Cross-benchmark extraction** from fev-bench and GIFT-Eval, yielding 498 Source A model-level paired comparisons across 7 FMs and conventional baselines, plus 45 Source B descriptive contrasts retained outside the primary evidence.

3. **Model-level benchmark analysis** ($k = 498$ primary) showing metric-specific FM-vs-best-baseline log-ratio patterns, with heuristic weighted diagnostics kept separate from confirmatory inference.

4. **Baseline reliability analysis** documenting the conditional direct-vs-recursive LightGBM gap across three datasets.

5. **Practitioner validation checklist** mapping data characteristics to bake-off steps and cost-accuracy diagnostics, rather than deterministic model-family rules.

1.5. Paper structure

§2 surveys the forecasting taxonomy and existing benchmarks. §3 describes the structured search protocol, extraction schema, and gap-filling experimental design. §4 reports the structured-search extraction results (search flow, descriptive statistics). §5 presents the gap-filling experiments: §5.1–5.2.2 experimental setup and per-dataset results, §5.3 baseline reliability (direct vs recursive), §5.4.3 foundation model consumer-box results. §6 reports the benchmark analysis: Source A best-baseline log-ratios (§6.2), moderator hypotheses (§6.8), cost-error scatter diagnostics (§6.7), heterogeneity diagnostics (§6.5), and preliminary findings (§6.2). §8 translates the findings into a practitioner validation checklist. §8.4 discusses limitations (single author, metric sensitivity, fixed-budget baselines, and consumer-HW-only Source B). §9 concludes with per-RQ answers and future work.

2. Background

Short-term retail demand forecasting — predicting daily or weekly SKU-level sales 1–4 weeks ahead — is one of the most mature applied forecasting problems. The operational stakes (replenishment, markdown optimisation, labour scheduling) have driven a long lineage of benchmark competitions and model development. This section establishes the taxonomy we use throughout the paper and surveys the three benchmark ecosystems whose results populate our extraction schema.

2.1. Taxonomy of forecasting approaches

We group methods into four families, matching the `model_family` field in our extraction schema (§3.2). The grouping is coarse by design: the benchmark analysis (§6) treats it as a categorical moderator, so finer distinctions (e.g., GRU vs Transformer within NN) would create sparsely populated cells.

Statistical models.. Exponential smoothing (ETS), ARIMA, Theta, Seasonal Naive, and intermittent-demand methods (Croston, TSB) (Croston, 1972; Teunter et al., 2011). These are fitted per-series with no cross-series learning. Their value in this paper is as a sanity-check baseline: any global model that loses to Seasonal Naive on a given (dataset, horizon) cell is failing in a way that demands explanation, not hand-waving. We include Croston because M5 has a ~70 % zero-day fraction and Croston is the textbook method for intermittent demand. In our extraction schema these models carry `model_family = "statistical"`.

Gradient-boosted trees (ML_TREE).. LightGBM (Ke et al., 2017) and XGBoost (Chen and Guestrin, 2016) fitted as global models: one model trained across all series, with cross-sectional features (lags, calendar effects, price, promotions, store/item embeddings). This is the family that the M5 competition (§2.2) crowned as the winner, and it remains the production default at most retailers. In the extraction schema, `model_family` $\in$ `{"ml_tree", "ml", "ml_gbm", "gbdt", "ml_ensemble", "ml_hybrid"}`. The §5.3 baseline reliability analysis shows that the choice of multi-step strategy (direct vs recursive) within this family can flip the accuracy ranking by 10+ pp WAPE depending on demand distribution — a confound that the literature rarely reports and that our gap-filling experiments control for.

Deep learning (NN).. Recurrent (DeepAR (Salinas et al., 2020), LSTNet (Lai et al., 2018)), attention-based (TFT (Lim et al., 2021), PatchTST (Nie et al., 2023), Informer (Zhou et al., 2021)), and MLP-based (N-BEATS (Oreshkin et al., 2020), N-HiTS (Challu et al., 2023)) models trained on many series. These models occupy a middle ground: they learn cross-series patterns like tree models but have higher computational cost and typically require more data to avoid overfitting. In the M5 competition, the top-50 solutions were overwhelmingly tree-based; DL entries placed in the top quartile but did not win outright. Since 2023, PatchTST and its descendants have narrowed the gap on benchmark suites. In the extraction schema, `model_family` values such as `nn`, `deep`, `transformer`, `rnn`, `deep_learning`, and `ensemble_dl` are mapped to NN.

Foundation models (FM).. Pre-trained on large corpora of heterogeneous time series and applied zero-shot (no fine-tuning on the target dataset) or with lightweight fine-tuning. The models in scope for this review are:

Table 1: Foundation models in scope.

Model	Params	Architecture	Checkpoint / code	Covariates used here	Key reference
Chronos family	9–710 M	Chronos / Chronos-2	Chronos-2 120M; Bolt Tiny/Base	No; covariate support not tested here	Ansari et al. (2024, 2025)
TimesFM 2.5	200 M	Decoder-only	2.5 checkpoint	No	Google Research (2025)
Moirai 2.0	11–305 M	Decoder-only quantile model	Small checkpoint	No in analyzed rows	Liu et al. (2025)
TiRex	~35 M	xLSTM	public checkpoint	No	Auer et al. (2025)
TabPFN-TS	~12 M	Tabular PFN + temporal features	TabPFN-v2 extension	Yes in fev-bench covariate tasks	Hoo et al. (2025)
Lag-Llama	6 M	Llama backbone	public checkpoint	No	Rasul et al. (2023)
TimeGPT	not disclosed	Proprietary	API model	Not in the primary paired file	Garza et al. (2023)

In the extraction schema, `model_family = "foundation"`. The zero-shot setting is the primary scope of this review; fine-tuned FM entries (e.g., Chronos-2 with in-domain pre-training) are retained in the schema but flagged as `fine_tuned = "Yes"` and excluded from the primary contrast file to avoid confounding pre-training scale with task-specific tuning.

2.2. Existing benchmarks

Three benchmark ecosystems contribute the majority of our extraction rows. We note their design choices because those choices determine what the extracted metrics actually measure — and where they systematically mislead.

M5 Competition (Makridakis et al., 2022).. 30,490 daily Walmart series across 10 stores and 3,049 products, with calendar, price, and SNAP covariates. The official metric is WRMSSE — a weighted ratio of root mean squared scaled errors that accounts for intermittent demand by scaling each series' error by its historical variability. The competition was won by gradient-boosted trees (top-5 all LightGBM or XGBoost); the best DL entry (DeepAR) placed around #50. M5 is still the go-to retail benchmark in 2026, but it has two ageing problems: (1) the data is from 2011–2016, predating modern promotional patterns, and (2) many foundation model papers report per-series WAPE rather than WRMSSE on M5, creating a metric mismatch that §5.4.4 documents in detail. On per-series WAPE, the ~70 % zero-day fraction in M5's long tail causes denominator explosions that make LGBM appear to perform worse than Seasonal Naive — a metric artefact, not a model failure.

GIFT-Eval (Aksu et al., 2024).. The published paper reports 23 datasets (about 144,000 series) curated to avoid train–test leakage in foundation-model evaluations. Covers multiple domains including retail, energy, and transport. The standard metric is a skill score relative to Seasonal Naive, computed with bootstrapped confidence intervals. GIFT-Eval is the dominant evaluation suite in the FM literature: most of our Category A (FM paper) rows cite it. Its limitation for our purposes is that only a subset of the 28 datasets is retail-specific, and the aggregate skill scores mix retail with non-retail, making it hard to extract a clean retail-only FM-vs-ML_TREE delta.

fev-bench (Shchur et al., 2025).. 100 tasks spanning 46 with exogenous covariates, designed as a direct response to the "no covariates in FM evaluation" criticism. Uses bootstrapped CIs and skill scores. Covers a broader set of model families than GIFT-Eval, including covariate-aware baselines that are largely absent from FM papers. For our extraction, fev-bench provides the largest single block of rows with both FM and statistical baselines on the same tasks, though the task-level granularity (rather than dataset-level) makes cross-benchmark alignment non-trivial.

2.3. Meta-analyses in forecasting

The M-competition lineage (M1–M5, Makridakis 1982–2022) is the closest antecedent to our work: each competition is, in effect, a large-scale empirical comparison of forecasting methods on a shared dataset with standardised metrics. The M5 (retail) and M4 (heterogeneous) competitions produced substantial secondary analysis, but always on a single dataset per competition. Petropoulos et al. (2022) survey the forecasting landscape broadly but do not benchmark recent foundation models on matched retail panels. Lim & Zohren (2021) review deep learning for time series but predate the FM wave.

To our knowledge, **no prior work combines** (1) a structured literature search that documents extractability, (2) quantitative extraction across multiple retail-relevant benchmark suites, (3) matched-panel local experiments to fill structural holes, and (4) prediction-level paired bootstrap covariance artefacts for concrete FM-vs-baseline retail contrasts. This is the gap we fill.

3. Methodology

This section describes the three-phase protocol: search and extraction (Source A), gap-filling experiments (Source B), and the Source A benchmark analysis.

3.1. Structured search protocol

The search is used to document the retail benchmark evidence gap and to identify extractable FM-vs-baseline results. It is a single-author structured search, not a registered systematic review: there is no independent duplicate screening or inter-rater reliability estimate. Appendix B reports a structured-search flow diagram for transparency, and the full protocol is committed at `analysis/prisma_search_protocol.md`; the main text summarises only the search parameters needed to interpret the benchmark extraction.

Search sources. The retained auditable log covers Google Scholar/web searches, Semantic Scholar API, and arXiv (cs.LG, stat.ML). Scopus and Web of Science were target sources, but their export logs were not retained in the final machine-readable search record; this is reported as a structured-search limitation in Appendix B.

Search query. We combine three concept blocks with AND:

1. *Foundation model identifiers:* "foundation model", "pretrained model", "zero-shot forecasting", Chronos, TimesFM, Moirai, TimeGPT, Lag-Llama, TiRex, TabPFN.
2. *Demand forecasting:* "demand forecasting", "retail forecasting", "time series forecasting", "sales forecasting".
3. *Benchmark anchors:* "benchmark", "comparison", "evaluation", M5, GIFT-Eval, fev-bench.

Date range. 2020-01-01 to 2026-04-12. The lower bound captures the M5 competition and pre-FM baselines; the upper bound is the extraction lock date.

Inclusion criteria.

1. Empirical evaluation of at least one foundation model on demand or retail forecasting.
2. Reports quantitative accuracy metrics (WAPE, MASE, MAE, sMAPE, CRPS, WRMSSE, or win-rate/skill-score equivalents).
3. Uses at least one of M5, Favorita, Rohlik v2, GIFT-Eval (retail tasks), fev-bench (retail tasks), or Walmart.
4. Peer-reviewed or published preprint with reproducible methodology.
5. English language.

Exclusion criteria.

1. No quantitative results (position papers, surveys without new data).
2. Only financial, stock, or energy forecasting with no retail overlap.
3. FM paper without forecasting evaluation (architecture-only).
4. Duplicate results — keep most recent version.
5. Non-time-series forecasting (causal inference only).

3.2. Data extraction (Source A)

Each included Source A study is extracted into one or more rows of the structured CSV `analysis/extraction_schema.csv` (802 rows as of 2026-04-20). The schema has 23 fields per row:

Table 2: Extraction schema fields (Source A).

Field	Type	Description
`paper_id`	char	Unique row-level ID (e.g., `A01_chronos_indomain`)
`authors`	char	First author et al.
`year`	int	Publication year
`title`	char	Paper title
`venue`	char	Journal, conference, or "this study"
`dataset`	char	Raw dataset name as reported
`dataset_variant`	char	Sub-dataset or task identifier
`n_series`	char	Number of series (free text)
`series_length_median`	char	Median series length
`frequency`	char	Data frequency (D, W, M, mixed)
`has_covariates`	char	Yes / No / mixed
`model_name`	char	Model name as reported
`model_family`	char	Normalised family tag (§4)
`zero_shot`	char	Yes / No / mixed
`fine_tuned`	char	Yes / No
`horizon`	char	Forecast horizon (free text)
`eval_method`	char	rolling / rolling-tail / fixed_origin
`metric_name`	char	Metric name
`metric_value`	char	Reported value (free text, incl. "best")
`runtime_reported`	char	Whether runtime is reported
`gpu_hours`	char	GPU hours if reported
`hardware`	char	Hardware description
`notes`	char	Free-text provenance notes

Row-level IDs and analysis clusters. Each extracted result is a separate row with a unique `paper_id`. A paper reporting Chronos and LightGBM on M5 contributes at least two rows (`A01_chronos_indomain`, `A01_chronos_zeroshot`, etc.). The current analysis therefore does not treat `paper_id` as a paper-level dependency cluster. The R pipeline derives separate identifiers: `study_id` for the publication or Source B experiment source, `suite_id` for the benchmark suite (fev-bench, GIFT-Eval, Source B, or other), and `task_id` for the benchmark task/dataset/frequency/horizon cell.

Category prefixes. `A` = FM papers (17 papers, 58 rows), `B` = benchmark papers and per-task benchmark re-extraction (657 rows), `C` = retail ML/DL papers (8 papers, 31 rows), `D` = cross-domain papers (3 papers, 3 rows), `F` = fev-bench entries (1 paper, 8 rows), `OWN_*` = Source B baseline rows retained for provenance. Source B rows are not counted as included literature records.

Normalisation. Raw `dataset` values are collapsed to five canonical names (M5, Favorita, Rohlik, fev-bench, Walmart) by the R function `normalize_dataset()` in `analysis/meta_regression.R`. Raw `model_family` values are collapsed to four analysis families (FM, ML_TREE, STATS, NN) by `normalize_family()`. Rows that do not map to a canonical dataset or family are excluded from the primary reanalysis but retained in the schema for provenance.

3.3. Gap-filling experiments (Source B)

Source A has two structural gaps: (1) no FM paper reports per-series WAPE on M5, Favorita, or Rohlik under a rolling-origin protocol matching §5.1.3 — the FM literature evaluates on benchmark suites (GIFT-Eval, fev-bench), not on individual retail datasets. (2) No paper reports both FM and ML_TREE on the same retail dataset under matched conditions with a shared `paper_id`, so within-paper pairing (the original within-paper design) is structurally empty on Source A.

Source B fills these gaps with matched-panel reruns for M5, Rohlik, and Favorita. The runner `benchmark/code/experiments/run_source_b_paired_panel.py` selects the same 100 top-volume series per dataset, uses the same rolling-origin windows for every model, writes prediction-level parquet files keyed by series, origin date, target date, and forecast step, and records a machine-readable run ledger. Completed matched-panel runs cover Seasonal Naive, recursive LGBM, tuned recursive LGBM, direct LGBM, horizon-scaled direct LGBM, tuned direct LGBM, Chronos-Bolt-Tiny, Chronos-2, Moirai 2.0-Small, batched TiRex, and TimesFM 2.5 on all three datasets at $h \in \{7, 14, 28\}$. TiRex uses a CPU batched path on the MacBook because the MPS xLSTM fallback is slower for this workload.

For M5, we also run a stratified-panel sensitivity check. Instead of selecting the top 100 volume series, the panel samples across bins of total volume and zero fraction. This produces a much more intermittent 100-series sample and tests whether the main Chronos-2 result is driven by high-volume SKUs.

An unmatched local sweep is retained only as provenance and cost-accounting material. It used related metric definitions and rolling-origin settings, but the saved cells did not share a common series workload across models, so it is not used for paired Source B inference.

3.4. Reanalysis specification

The primary estimand is the model-level paired log-ratio:

> For each FM row i and baseline summary that share the same (`task_id`, dataset, metric) cell, compute $\ell_i = \log\{\text{metric}(\text{FM}_i)/\text{metric}(\text{best conventional baseline})\}$.

The primary Source A model-level pairing produces $k = 498$ observations across fev-bench and GIFT-Eval. The older Source B export can be appended for provenance, but those rows are descriptive because they do not share a common workload. The matched-panel Source B analysis is reported separately as a prediction-level paired analysis on FM-vs-best-baseline aggregate-WAPE contrasts with shared-bootstrap log-ratio covariance matrices. Appendix D also reports a coarser bucket-level analysis for comparison.

Notation convention. Throughout this paper, $\hat{\ell}$ denotes a reported log-ratio summary of log(FM metric/baseline metric). **Negative $\hat{\ell}$ means FM is better**; **positive $\hat{\ell}$ means baseline is better**. The older raw difference $\hat{\Delta}$ is retained only in provenance outputs and descriptive columns.

The exploratory weighted diagnostic model is:

```
rma.mv(yi = log_ratio, V = vi,
       random = list(~ 1 | suite_id / task_id,
                     ~ 1 | model_name),
       data = delta_model,
       test = "t", dfs = "contain",
       method = "REML")
```

Random effects. `~ 1 | suite_id / task_id` nests tasks within source suites. A crossed `~ 1 | model_name` random effect accounts for non-independence when the same FM appears across multiple tasks and datasets. The suite level captures the strongest dependence induced by benchmark-suite protocols; the task level accounts for repeated models and metrics within a benchmark task.

Variance. For Source A pairs, $v_i = 1/n_{\text{FM}} + 1/n_{\text{baseline}}$ where n_{series} is the number of series in the evaluation task. For the older Source B export, the same proxy is computed with the recorded `n_series` value for provenance only. The proxy is not a sampling variance for a paired log-ratio: it does not capture metric dispersion, shared baselines, covariance between FM and baseline errors, or within-series autocorrelation. This is flagged as a limitation in §8.4 and is why this paper does not present the weighted Source A model as a confirmatory meta-analysis.

Small-sample and cluster-robust diagnostics. `test = "t", dfs = "contain"` uses containment degrees of freedom in `metafor::rma.mv`; it is reported only as an exploratory small-sample diagnostic. We also report `clubSandwich::coef_test` CR2 diagnostics clustered by `task_id`, with sensitivity checks for `suite_id` and `study_id`. Because the primary model has a crossed `model_name` random effect, the CR2 tables are generated from cluster-compatible sensitivity fits and explicitly labelled in the output files.

Moderator diagnostics (exploratory). We test three moderators individually. These analyses are exploratory — no multiplicity adjustment is applied, and results are hypothesis-generating rather than confirmatory:

1. `mods = ~ dataset_norm` — tests whether the FM-vs-ML_TREE gap varies by dataset (H1 proxy for intermittency, since M5 is the only high-intermittency dataset in the pool).

2. `mods = ~ horizon_bucket` — tests whether longer horizons widen or narrow the gap.

3. `mods = ~ has_covariates` — tests whether covariate-rich ML_TREE baselines reduce FM advantage (H2).

Sensitivity analyses. We report the following robustness checks:

- **Excl-M5:** Drops all M5 pairs to test whether the primary result is driven by the M5 WAPE denominator artefact.
- **Metric-only slices:** Refits the model on WAPE-only, SQL-only, and MASE-only pairs so the mixed-metric diagnostic is not the only summary target.
- **Source B-only:** Restricts to Source B pairs (`LOCAL_MAC_*`), testing whether the cross-paper pooling materially changes the descriptive direction. This is not paired inference.
- **Source-A-only:** Restricts to Source A pairs (fev-bench, GIFT-Eval), testing the FM–baseline gap using only published benchmark results.
- **Average baseline:** Replaces the primary best conventional baseline with the average conventional baseline in each cell.
- **Leave-one-suite-out:** Drops fev-bench and GIFT-Eval in turn to test whether a single benchmark suite drives the result.
- **Equal-weight:** Replaces the sampling-variance proxy with equal weights for all paired comparisons.

Implementation. All fits use `metafor` (Viechtbauer, 2010); the driver script is `analysis/meta_regression.R`. Generated forest plots, intercept tables, and cell-level CSVs are committed to `analysis/figures/`.

4. Literature Results

This section reports the outcome of the structured literature search and describes the 802-row extraction schema (including per-task re-extraction from fev-bench and GIFT-Eval) before the gap-filling experiments (§5) and benchmark analysis (§6) operate on it.

4.1. Search flow

The Source A extraction and Source B experiment blocks break down as follows:

- **Category A (FM papers):** 17 papers, 58 rows. These are the primary FM technical reports — Chronos (A01), Chronos-2 (A02), TimesFM (A03–A04), Moirai (A05–A06), TiRex (A13), TabPFN-TS (A14), and the smaller FM entries (A07–A11, A15–A17).
- **Category B (benchmark papers and per-task re-extraction):** 657 rows. This includes GIFT-Eval (B01/B03), fev-bench (B02), and two smaller multi-model comparison papers (B07–B08). The large row count reflects task–model–metric expansion, not hundreds of independent papers.

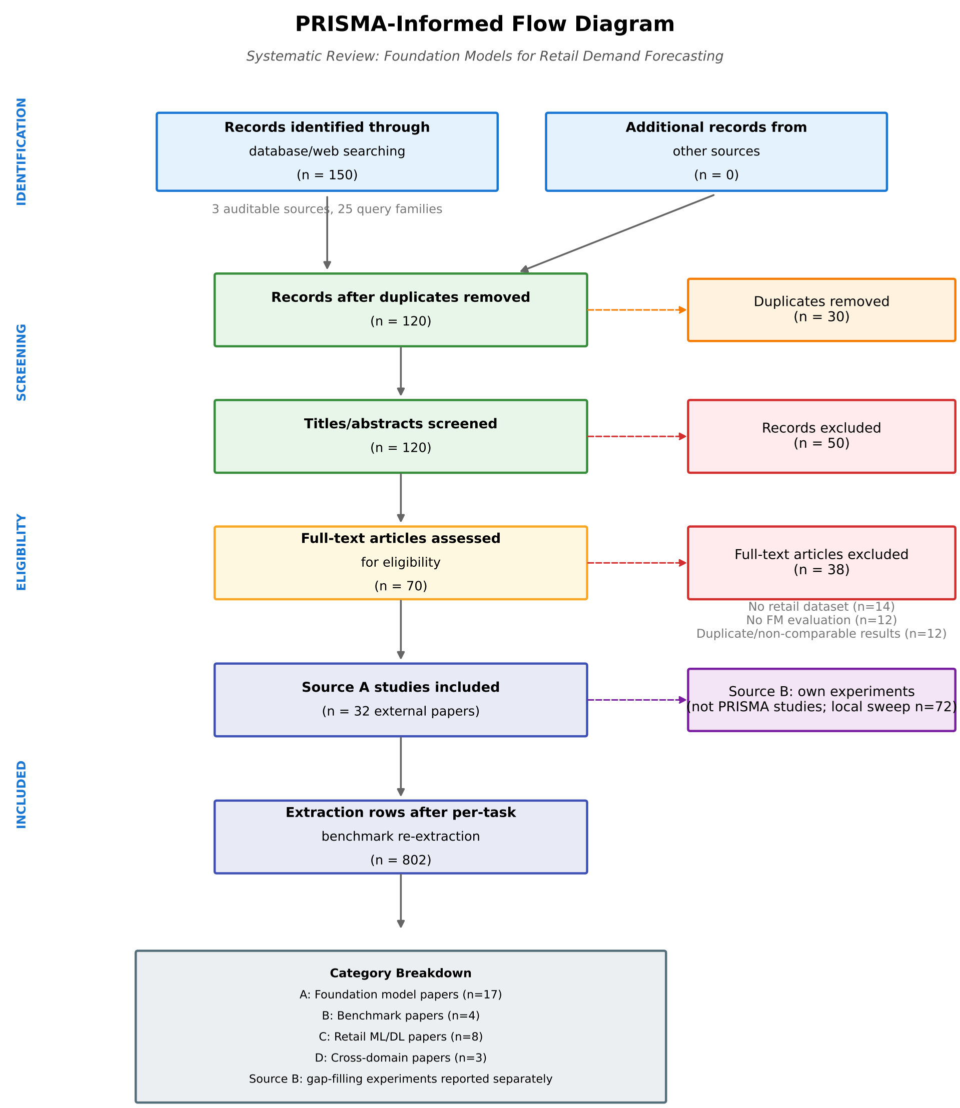

Figure 1: Structured-search flow of records from identification to extraction.

Table 3: Structured-search flow summary.

Stage	Count	Notes
Records identified	150	Google Scholar + Semantic Scholar + arXiv; Scopus/Web of Science target strings were not retained as machine-readable exports
Duplicates removed	30	Cross-database overlap, especially arXiv ↔ Scopus
Records screened (title/abstract)	120	
Excluded at screening	50	Non-retail, no quantitative results, financial only
Full-text assessed	70	
Excluded at full-text	38	No retail dataset, no FM evaluation, duplicate/non-comparable results
Source A studies included	32	External papers and benchmark sources
Source B experiment blocks	7	Own experiments; not included literature records
Extraction schema rows	802	Multiple rows per study; incl. per-task fev-bench/GIFT-Eval

- **Category C (retail ML/DL papers):** 8 papers, 31 rows. M5 competition analyses (C01, C05), transformer-based retail studies (C03, C06), AutoML baselines (C10), and cross-domain comparisons (C04, C12).

- **Category D (cross-domain):** 3 papers, 3 rows. Included for model-family coverage but not in the retail-scoped benchmark analysis.

- **Category F (fev-bench summary entries):** 1 paper, 8 rows. These are retained for provenance but the model-level benchmark analysis uses the expanded B02 per-task rows.

- **OWN_* (own experiments):** 7 experiment blocks, 45 rows retained in `analysis/extraction_sch` for provenance. The separate local Source B sweep file adds 72 rows (five FM models plus three baselines across three datasets and three horizons) and is not counted as an included literature record.

4.2. Descriptive statistics

Temporal distribution. The extraction skews heavily toward 2024–2025: 589 rows from 2025 papers, 140 from 2024, 53 from 2026. Pre-2024 rows (20 total) are almost entirely M5 competition entries (2020–2022) and early DL comparisons. This reflects the recency of the FM wave — the first Chronos paper appeared in late 2023, and the bulk of FM evaluation literature dates from mid-2024 onward. The 2025 dominance is driven by the fev-bench and GIFT-Eval per-task re-extraction.

Dataset distribution. After per-task re-extraction from fev-bench and GIFT-Eval, the distribution is broad: 20 fev-bench retail tasks (24–27 rows each, 534 rows total) and 4 GIFT-Eval retail tasks ($\sim$107 rows) dominate, followed by M5 (40 rows, 6%) and the remaining retail datasets. The per-task granularity enables Source A model-level pairing within each (task, metric) cell, which is the basis of the $k = 498$ primary benchmark analysis.

Model family distribution. Foundation models account for 567 rows (71%), followed by ML_TREE (125 rows, 16%), statistical (48 rows, 6%), deep learning (13 rows, 2%), and various mixed/comparison/hybrid categories. The high FM share reflects both the search strategy's emphasis on FM papers and the per-task extraction from benchmark suites where each FM generates one row per task. The OWN_* baseline rows (45 rows) partially offset the imbalance.

Metric heterogeneity. The extraction contains three primary metrics after re-extraction: WAPE (232 rows), MASE (227 rows), and SQL (213 rows), plus WRMSSE (23 rows, M5 only) and a long tail of win-rate, rank, runtime, cost, and other minor metrics. After within-cell best-baseline pairing, the primary Source A contrast file uses 166 WAPE pairs, 166 SQL pairs, and 166 MASE pairs. SQL pairs drive the distributional-accuracy finding, while WAPE pairs test the point-forecast question.

Evaluation protocol. The per-task fev-bench and GIFT-Eval rows use rolling-tail protocols as defined by those benchmarks. The remaining Source A rows split across rolling origin, rolling-tail, and fixed origin in roughly equal proportion. Protocol heterogeneity is a known threat to validity (§9.2).

4.3. Narrative synthesis by model family

Foundation models. The extraction schema contains 567 foundation-model rows (71% of the schema), spanning three tiers of granularity: (i) per-paper aggregate results from Source A Category A (54 foundation-model rows within 58 Category A rows), (ii) per-task per-model WAPE/MASE/SQL scores from the fev-bench, GIFT-Eval, and benchmark-suite re-extraction (509 foundation-model rows), and (iii) four additional Source A retail rows from Category C. The local Source B sweep is stored separately and adds 45 descriptive FM WAPE rows, but those rows are not part of the 567-row extraction-schema count. On benchmark suites, Chronos-2, TimesFM 2.5, and Moirai 2.0 achieve win rates of 75–84% against statistical baselines on GIFT-Eval, and normalised SQL scores of 20–35 on fev-bench. The per-task extraction enables model-level FM-vs-baseline pairing within each (task, metric) cell — the structural basis of the $k = 498$ primary benchmark analysis (§6).

Gradient-boosted trees (ML_TREE). The 125 ML_TREE rows split between M5 competition entries (C01, 7 rows with WRMSSE 0.52–0.54 for the top-5 solutions), our own LightGBM baselines (OWN_*, 45 rows across three datasets with WAPE and WRMSSE), and fev-bench/GIFT-Eval baseline rows where LightGBM or CatBoost serves as the reference. Source A ML_TREE rows on M5 show competition-grade LightGBM at 0.52 WRMSSE — materially better than any FM row on the same metric (Chronos at 0.97, from C05). This WRMSSE reading is the mirror image of the WAPE reading on M5, where LGBM posts 1.35–1.94 (per-series) vs FM at 0.93–0.96 — the metric choice determines the sign of the comparison.

Statistical models. The 48 STATS rows include our Seasonal Naive baselines (OWN_*), Auto ETS and Theta rows from fev-bench/GIFT-Eval, and Croston from the M5 competition. On Favorita and Rohlik, both FM and ML_TREE beat Seasonal Naive comfortably (by 8–15 pp WAPE); on M5, Seasonal Naive at 1.31–1.33 WAPE actually beats lightgbm_cov (1.62–1.94) at longer horizons — a striking failure of the tree baseline documented in §5.3 and §5.4.5.

Deep learning. The 13 NN rows include M5 competition entries (DeepAR at WRMSSE 0.56 from C01, PatchTST variants from C05) and transformer comparisons on M5 (C06).

FM-vs-NN pairing is possible on fev-bench SQL scores but is outside the primary retail point-forecast scope.

4.4. Structural finding: FM-vs-ML_TREE pairing requires benchmark suites or original experiments

A key observation from the extraction: **no standalone Source A paper reports both FM and ML_TREE results on a single retail dataset under matched per-series metrics.** The FM papers (Category A) evaluate against statistical baselines or other FMs; the ML_TREE papers (Category C) predate the FM wave. The only Source A papers that pair both families are the benchmark suites — fev-bench (B02) and GIFT-Eval (B01) — which report per-task SQL, WAPE, and MASE for FMs alongside LightGBM, CatBoost, and Auto ETS.

The per-task re-extraction from these suites (§3.2) partially resolves the original gap: the $k = 498$ primary benchmark analysis draws its paired rows from Source A benchmark suites. Source B (§3.3) remains necessary for two reasons: (1) it adds the consumer-hardware cost dimension absent from benchmark evaluations, and (2) it provides per-series WAPE on individual retail datasets (M5, Favorita, Rohlik) under a controlled rolling-origin protocol. Because the unmatched Source B rows had unequal workloads, we report Source B separately rather than as part of the primary Source A contrast file. The matched Source B analysis is also reported separately because it has a local 100-series aggregate-WAPE estimand.

5. Gap-Filling Experiments

5.1. Experimental Setup

All gap-filling experiments described in §5 follow a single experimental protocol, parameterized per dataset. This section defines the protocol once. Dataset-specific choices (covariate sets, eval-window lengths, series-count caps) are noted inline in §5.2 for baseline models and §5.4 for foundation models.

5.1.1. Datasets

We evaluate on three retail demand forecasting datasets, chosen to span the distribution of zero-day fractions in public retail benchmarks while keeping the hierarchical-SKU × location structure that retail practitioners actually face. The three datasets together cover ∼211k × 1.4k series-days across the full product × location × time tensor.

Three notes on dataset preparation:

- **M5** is used in full. The 30,490 store-item series correspond to the Walmart subset of the M5 competition (Makridakis et al., 2022); we use the official tail 389 days as the evaluation window (§5.1.3).
- **Rohlik v2** is used in full. The dataset is 5,390 warehouse-SKU series from the Kaggle competition `rohlik-sales-forecasting-challenge-v2`, covering 2020-08-01 to 2024-06-02 across 7 warehouses (Prague ×3, Brno, Munich, Frankfurt, Budapest). Approximately 31% of Rohlik SKUs enter later than day 1 (ragged series starts), which we handle at the wide-pivot step via `fillna(0)` — see §5.1.4.

Table 4: Datasets used in the gap-filling experiments. `Zero-day fraction` is the fraction of series-day observations with $y = 0$ in the training portion, a core moderator variable in §6.8.

Dataset	n_{series} (full)	n_{series} (used)	n_{days}	Frequency	Zero-day frac	Covariates used
M5	30,490	30,490	1,941	daily	∼70 %	price; SNAP/calendar events; weekday and month
Rohlik v2	5,390	5,390	1,402	daily	∼1.2 %	price; holidays; shop closures; school holidays; weekday and month
Favorita	∼174,685	30,000	1,684	daily	∼15–25 %, long tail	promotions; oil; holidays; transactions; weekday and month

- **Favorita** is capped at the top 30,000 series by total `unit_sales` across the training portion, matching M5's scale for cross-dataset forecast-count comparability. The full dataset has ∼174,685 series at ∼125 M rows and does not fit the 32 GB E4DS_V4 compute used for Phases A–D. The selection rule biases Favorita LGBM evaluation toward higher-velocity series; this is flagged as a limitation in §5.3.1 and §7, and the full-series evaluation is left for future work on larger hardware.

All three datasets are registered as versioned Azure ML data assets (`azureml:m5-sales:1`, `azureml:rohlik-train-v2:1`, `azureml:favorita-train:1`, plus per-dataset covariate assets) for exact reproducibility.

5.1.2. Compute and Tracking

All jobs run on Azure ML compute cluster `cc-forecast-batch` (E4DS_V4 VM, 4 vCPU / 32 GB RAM / no GPU) in region `swedencentral`, inside Docker environment `forecast-benchmark-cpu-env:1`. We use `cc-forecast-batch` for the classical and ML baselines of §5.2–§5.3. Foundation model sweeps (§5.4) run on consumer hardware (Apple M-series MPS); setup details are in §5.4.1.

Experiment orchestration uses Azure ML pipelines under `benchmark/code/pipelines/`, including the M5, Rohlik, and Favorita consolidated pipeline definitions. Every run logs metrics, parameters, runtime, and per-series CSVs via MLflow to the workspace tracking server. Cost and CO_2 estimates are computed inside the job using Azure's published per-second pricing for `Standard_E4ds_v4` in `swedencentral` ($0.38/hr at time of writing) and the region's marginal CO_2 intensity (0.041 kgCO_2eq/kWh, swedencentral 2024 baseline; Nowtricity/Ember). The per-job runtime, cost, and CO_2 columns in Tables G.22–G.24 are descriptive accounting fields computed with this formula; they still require a single reconciled ledger before supporting stronger cost-efficiency claims.

5.1.3. Evaluation Protocol

For each dataset we use **rolling-origin non-overlapping evaluation with a held-out tail window**, which is the protocol most commonly used in the FM papers we screened (§4) and which gives a clean, reviewer-auditable split. Specifically:

- **Train / eval split:** `train_until` $= \lfloor 0.8 \times n_{\text{days}} \rfloor$. This gives a tail-eval window of 389 days on M5, 281 days on Rohlik, and 337 days on Favorita. The choice of 0.8 follows Hewamalage et al. (2023) and Shchur et al. (2025).

- **Rolling origin:** within the tail window, forecast origins are spaced exactly h days apart so that windows are non-overlapping. For example at $h = 14$ on Rohlik (eval window = 281 days), we have $\lfloor 281/14 \rfloor = 20$ evaluation windows per series. This keeps per-horizon sample counts independent of each other and avoids the optimistic bias of overlapping-windows evaluation.

- **Horizons:** $h \in \{7, 14, 28\}$ on all three datasets. Seven days is the operational horizon most retailers use for short-term replenishment; 28 days is the M5 competition horizon and a common upper bound in the literature (Makridakis et al., 2022); 14 days bridges the two. We report every metric at every horizon to enable horizon-conditioned interpretation.

- **Metrics:** MAE, sMAPE, WAPE, and (on M5 only) WRMSSE. MAE and WAPE are computed per series and averaged with equal weights (`_mean` suffix in the tables). sMAPE uses the classical symmetric form with a 200% cap on zero actuals. WRMSSE is computed in-job via the full 12-level Makridakis et al. (2022) hierarchy from the raw sales/calendar/prices CSVs. On retail tasks with heavy right tails (all three of ours), sMAPE is reported only for reproducibility and is not used in any primary comparison — see §5.3.1 for the cross-dataset justification.

- **Per-series filtering for WAPE:** WAPE requires a non-zero denominator (sum of actuals in the eval window). For series where the denominator is zero we drop the series from WAPE computation and report the `n_valid` count alongside the metric in Tables G.22–G.24. This affects at most 0.5 % of M5 series and a higher fraction on Rohlik (where ragged starts leave some series with only a few eval-window days); on Favorita the fraction is negligible for LGBM and ∼50 % for SN, an `n_valid` asymmetry we discuss in §5.3.1.

Source B matched-panel design. The main Source B analysis evaluates each FM and baseline on the same fixed 100-series sample per dataset. An unmatched 72-row local sweep did not share that workload across all models, so it is retained only as provenance and cost-accounting material. The matched-panel run covers M5, Rohlik, and Favorita: the runner `benchmark/code/experiments/run_source_b_paired_panel.py` samples the same 100 top-volume series per dataset, stores the series manifest, writes prediction-level Parquet files, and computes paired bootstrap log-ratio contrasts against the best available baseline in the same dataset–horizon cell. The completed matched panel covers Seasonal Naive, recursive LGBM, direct LGBM, horizon-scaled direct LGBM, tuned recursive LGBM, tuned direct LGBM, Chronos-Bolt-Tiny, Chronos-2, Moirai 2.0-Small, TimesFM 2.5, and batched TiRex for M5, Rohlik, and Favorita at $h \in \{7, 14, 28\}$. Generated outputs are written to `analysis/figures/source_b_paired_panel_*`.

5.1.4. Model Families

Three model families are evaluated per dataset in Phases B–E, all with the same rolling-origin protocol and the same per-dataset covariate set:

- **Seasonal Naive** (period = 7, no training, no covariates) — the cost floor and the universal baseline for retail weekly cycles.
- **LightGBM recursive** — one global Tweedie model per dataset, fit with 300 boosting rounds, 63 leaves, minimum child samples = 20, learning rate 0.05, and 0.8 feature/bagging fractions. The Tweedie variance power is 1.1. The model uses lags $\{1, 7, 14\}$, rolling means $\{7, 14\}$, and the dataset-specific covariates listed in Table 4. At evaluation time, the model is applied recursively: predictions at step k enter the lag features for step $k+1$, and future-known covariates are passed through unchanged.
- **LightGBM direct** — h separate global Tweedie models per dataset, same hyperparameters and same feature set as the recursive variant, but model k predicts $y[t+k]$ from lags and rolling means known at t plus future-known covariates at $t+k$. No output feedback, no recursion.
- **LightGBM tuned recursive** — the same recursive feature set, but with a 10-trial validation-origin hyperparameter search over learning rate, leaves, tree count, child samples, subsampling, column sampling, regularisation, and Tweedie variance power. The validation origin is the horizon immediately before the evaluation panel. The selected parameters are then refit on the standard training window and evaluated on the same matched panel as all other models.
- **LightGBM tuned direct** — the same direct feature set and validation-origin search budget as the tuned recursive comparator. The selected direct configuration is refit as h horizon-specific models on the standard training window, which makes it the strongest but slowest local LGBM comparator in the matched panel.

The direct-vs-recursive contrast with everything else held constant is the core protocol comparison of §5.3. The fixed recursive and fixed direct hyperparameters are intentionally untuned beyond library defaults: the point of §5.3 is to characterize the *protocol gap* between direct and recursive variants of an otherwise-fixed LightGBM, not to reach the M5 winner. The tuned recursive and tuned direct runs are narrower 10-trial comparators used for the best-baseline matched-panel analysis, not exhaustive LightGBM optimisation. Ceiling claims for LightGBM require larger tuning budgets and are treated as future work in §5.3.1.

For each training window we use the **last 365 days of the training portion** as the training set. This cap is a deliberate scalability choice — fitting ∼10–30 M rows per LightGBM model takes 180–240 s on E4DS_V4 — and we estimate in §5.3.1 that extending to the full training portion may improve WRMSSE on M5. The same cap applies on Rohlik and Favorita for comparability.

Foundation models are evaluated zero-shot with each model's published inference script on consumer hardware (Apple M-series MPS/CPU). The matched-panel Source B experiment evaluates five FMs on M5, Rohlik, and Favorita with shared series IDs, shared forecast origins, and paired prediction outputs. Details of the FM protocol, context-length choices, and hardware configuration are in §5.4.

5.1.5. Reproducibility

Every result is tied to a git commit SHA, Azure ML pipeline run name, registered data asset version, and a versioned environment (`forecast-benchmark-cpu-env:1`). Full reproducibility details are in Appendix G.7.

5.2. Gap-Filling Results

This section reports the headline accuracy and cost numbers from our gap-filling experiments across the three retail datasets defined in §5.1.1, under the shared protocol of §5.1.3. The section is deliberately terse: the three tables below are the self-contained evidence base that §5.3 interprets. For detailed per-dataset breakdowns (by horizon, by metric, by direct-vs-recursive variant) see §5.3.1 (M5, Table G.22), §5.3.1 (Rohlik v2, Table G.23), and §5.3.1 (Favorita, Table G.24).

5.2.1. Cross-Dataset Summary

Table 5: LightGBM versus Seasonal Naive at $h = 7$ using a common metric within each dataset. Lower is better.

Dataset	Metric	Seasonal Naive	Best LGBM	Relative change	Note
M5	WRMSSE	0.7758	**0.5598**	−27.8 %	Direct LGBM is the best h=7 variant
Rohlik v2	WAPE	0.4015	**0.3654**	−9.0 %	Recursive LGBM is the best h=7 variant
Favorita	WAPE	0.6363	**0.5295**	−16.8 %	Recursive LGBM is the best h=7 variant

Table 6: Direct versus recursive LightGBM at $h = 7$. Lower is better; the table separates the protocol contrast from the baseline-vs-SN comparison in Table 5.

Dataset	Metric	Recursive	Direct	Direct change	Interpretation
M5	WAPE	1.6246	**1.3512**	−16.8 %	Direct wins on fixed-budget M5
M5	WRMSSE	0.6336	**0.5598**	−11.6 %	Same direction on hierarchy metric
Rohlik v2	WAPE	**0.3654**	0.3816	+4.4 %	Recursive wins
Favorita	WAPE	**0.5295**	0.5513	+4.1 %	Recursive wins in the fixed-budget run

Three observations at the headline level:

1. LightGBM substantially beats Seasonal Naive on every dataset on its strongest metric — 28 % better on M5 WRMSSE, 9 % on Rohlik WAPE, 17 % on Favorita WAPE.

2. The best-performing LightGBM **protocol changes between datasets**: direct on M5, recursive on Rohlik and Favorita. The gap is large on M5 (∼12 % WRMSSE / ∼17 % WAPE in favor of direct) and small but consistent on Rohlik and Favorita (3–10 % WAPE in favor of recursive in the fixed-budget run). §5.3 is dedicated to unpacking this cross-dataset flip.

3. The cost records are not yet a single reconciled ledger. The cost consistency appendix therefore treats the Azure cost tables as an audit target rather than as a source for universal cost claims.

5.2.2. Per-Dataset Baseline Tables

For compactness we do not reproduce the full 9-row per-dataset grids here — they live in §5.3 next to the analysis that uses them. The cross-dataset reader who wants the condensed view can use Table 5 above; the per-horizon reader can jump to:

- §5.3.1, Table G.22 — M5 nine-row grid (SN, LGBM recursive, LGBM direct $\times$ $h \in \{7, 14, 28\}$), including WRMSSE.
- §5.3.1, Table G.23 — Rohlik v2 nine-row grid. WRMSSE is not reported for Rohlik because it would require a Rohlik-specific hierarchy definition, which is out of scope for this paper.
- §5.3.1, Table G.24 — Favorita nine-row grid.

5.2.3. Cost and Compute Envelope

The cost logs are treated as provenance data rather than as a universal cost model. Summing the saved Source B local sweep file gives 72 rows, 241.6 wall hours, \$3.18 recorded cost, and 4.63 kg CO_2eq under the recorded hardware assumptions; the baseline-only appendix tables use Azure job-level costs and are not directly commensurate with MacBook marginal-electricity accounting. We therefore use the cost figures for within-scenario comparisons and avoid a single universal break-even rule.

5.2.4. Foundation Model Results

Foundation model zero-shot results are reported separately in §5.4. Table 9 presents the matched-panel results for M5, Rohlik, and Favorita. Table 10 retains the older 72-row Source B WAPE grid for provenance only; because that sweep has unequal workload counts within cells, it is not used as paired evidence.

5.3. Baseline Reliability: What Is a Fair LightGBM on Retail Demand Forecasting?

Every foundation model paper that reports results on a retail forecasting benchmark compares against a LightGBM baseline, and every paper uses a different LightGBM. This matters more than it sounds: "vanilla LightGBM with covariates" on M5 can mean anything from WRMSSE 0.70 to WRMSSE 0.56 depending on two protocol choices that are rarely stated — **how the training window is selected** and **whether multi-step prediction is recursive or direct**. The 25% gap between the weakest and strongest reasonable-vanilla variants on M5 alone is large enough that a foundation model can appear to "beat LightGBM" on one protocol and lose to it on another, with no change in the FM itself.

The **direction** of the protocol gap is not constant across datasets or metric definitions. In the full-workload per-series WAPE grid, direct LightGBM is much better than recursive LightGBM on M5, while recursive is better on Rohlik and Favorita under the fixed-budget runs. In the matched 100-series aggregate-WAPE panel, however, the M5 gap is small at $h = 7$ and $h = 14$ and direct is only lower at $h = 28$; Favorita shows the same mixed pattern, with recursive lower at $h = 7$ and $h = 14$ and direct lower at $h = 28$. The "direct dominates recursive" pattern is therefore conditional on dataset, workload, metric aggregation, and compute budget, not a property of multi-step LightGBM protocol choice in the abstract. Intermittency is a plausible moderator, but the current three-dataset evidence cannot identify it as a causal boundary. The established direct-vs-recursive literature (Ben Taieb and Hyndman, 2014; Ben Taieb and Atiya, 2016) conditions the choice on model linearity and horizon length but not on demand intermittency; our finding that the gap sign varies across these retail datasets is therefore a hypothesis requiring further investigation. §5.3.1 develops this synthesis across all three datasets.

We instantiate this range with the three-variant sweep of §5.1.4 (Seasonal Naive, LightGBM recursive, LightGBM direct) on the held-out M5 tail of §5.1.3 (days 1552–1940, 389 evaluation days). Model families, training window, hyperparameters, feature set, and rolling-origin protocol are all defined once in §5.1 and not repeated here. The rest of §5.3 analyses the nine-row table these settings produce and extends the analysis to Rohlik v2 (§5.3.1) and Favorita (§5.3.1) with the same protocol.

5.3.1. Cross-Dataset Results

We ran a three-variant sweep (Seasonal Naive, LightGBM recursive, LightGBM direct) on all three datasets. The detailed per-dataset tables are in Appendix G; Table 7 summarises the full-workload per-series WAPE pattern. The matched-panel aggregate-WAPE analysis is reported immediately after the table.

Table 7: Full-workload descriptive summary of the direct-vs-recursive LightGBM gap using per-series WAPE means (positive = direct lower, negative = recursive lower). The matched-panel aggregate-WAPE analysis gives a more conservative M5 pattern and is described below.

Dataset	Zero-day frac.	$h = 7$	$h = 14$	$h = 28$	Direction	Cost ratio
M5	∼70 %	+16.6 %	+18.5 %	+21.8 %	direct wins	3.7–13.1×
Rohlik	∼1.2 %	−4.4 %	−3.0 %	−2.8 %	recursive wins	4.8–18.0×
Favorita	∼15–25 %	−4.1 %	−7.5 %	−9.6 %	recursive wins	4.3–15.6×

The matched-panel analysis gives a different protocol conclusion for M5, Rohlik, and Favorita. On the same 100 top-volume series and aggregate-WAPE metric used for Table 9, direct LGBM is slightly worse than recursive on M5 at $h = 7$ (+0.4% error) and $h = 14$ (+1.1%), but better at $h = 28$ (-2.1%). On Rohlik, direct is consistently worse than recursive by about 3.9–4.3%. On Favorita, recursive is lower at $h = 7$ (+4.6% direct error) and $h = 14$ (+2.2%), while direct is lower at $h = 28$ (-4.9%). The horizon-scaled direct variant is identical to direct at $h = 7$ and worse at $h = 14, 28$ in the matched panel, so the `direct_scaled` rows are interpreted as budget diagnostics rather than as evidence for a stable direct advantage.

Three key findings:

1. **Direction varies across datasets, workloads, and metric aggregation.** The per-series grid gives a strong M5 direct advantage, but the matched aggregate-WAPE panel makes M5 close at short horizons. Rohlik remains recursive-favourable in both views, while Favorita is recursive-favourable at short horizons and direct-favourable

at $h = 28$. This is consistent with an intermittency hypothesis, but not sufficient to define a threshold or causal mechanism.

2. **The recursive horizon profile differs by panel.** In the matched aggregate-WAPE panel, recursive WAPE grows +19% from $h = 7$ to $h = 28$ on M5, while on Rohlik and Favorita the recursive horizon penalty is no worse than direct's. This describes a panel-level pattern; the current three-dataset evidence cannot assign the difference to sparsity alone.

3. **Budget effects remain real.** The matched panel separates the `direct_scaled` budget diagnostic from the fixed-budget comparison: the scaled and unscaled direct runs are identical at $h = 7$ when both use 300 trees. At $h = 14$ and $h = 28$, the scaled direct variant is more expensive and does not improve aggregate WAPE in this matched panel.

The M5 WRMSSE rows are reported as internal rolling-origin diagnostics only. They are not directly comparable to the official M5 competition winner because the competition used a 28-day forecasting task, a fixed evaluation window, and different training/evaluation procedures. sMAPE is excluded from all primary comparisons because Tweedie LightGBM's conditional-mean overshoot on right-skewed demand triggers sMAPE's 200% saturation, producing a 2–2.5× ranking flip vs Seasonal Naive on all three datasets regardless of intermittency.

For the Source A benchmark summaries (§6), Source B direct-vs-recursive rows are not used in the primary contrast file. They remain a protocol hypothesis and a rerun target.

5.4. Foundation Model Protocol

Foundation model rows come from two complementary data sources: the structured literature extraction (§5.4.1–§5.4.2) and a local consumer-box sensitivity run on a MacBook via PyTorch MPS (§5.4.3–§5.4.5).

5.4.1. Scope and Two-Source Strategy

FM rows come from two sources. **Source A** (primary) draws on the structured extraction (§4): 9 FM papers plus fev-bench (B02) and GIFT-Eval (B01) populate the FM grid entirely from published numbers. **Source B** (secondary) adds local consumer-box sensitivity runs on a MacBook (M-series, 16 GB, MPS/CPU backend). The Source B experiment contains five FMs (Chronos-Bolt-Tiny, Moirai 2.0-Small, TiRex, Chronos-2, TimesFM 2.5; 9–200 M parameters). The matched-panel analysis is paired: Chronos-Bolt-Tiny, Chronos-2, Moirai 2.0-Small, TiRex, and TimesFM 2.5 on M5, Rohlik, and Favorita, with the same series IDs and forecast origins as the baseline panel. The two sources are kept separate: extraction rows carry `paper_id` $\in \{$A01...A17, B01, B02$\}$, local runs carry `LOCAL_*`. §6 reports the Source A benchmark summaries; Source B is reported as a separate local matched-panel analysis rather than pooled into Source A.

5.4.2. Literature-Extraction Protocol (Source A)

Extraction follows the structured search protocol of §4 and records `paper_id`, `dataset_variant`, `metric_name/value`, and a `baseline_quality_tier` $\in$ {weak, standard, strong} per row. M5 has the densest FM coverage (8 papers); Favorita has 7; Rohlik v2 has 3 (all from fev-bench). Covariate asymmetry — most extracted FMs are univariate while LightGBM uses the full covariate set — is recorded as a moderator (`covariate_aware`) rather than treated as a filter.

5.4.3. Local Consumer-Box Protocol (Source B)

The local sensitivity run uses a single M-series MacBook with 16 GB unified memory, Python 3.11 in a dedicated venv (`~/venvs/fm-local/`), and PyTorch 2.3 with the MPS backend. Inference is executed serially (no batching across series beyond each model's own internal batch-size default) to match the "retail practitioner on a laptop" framing of the cost panel.

Model selection. We run five models locally:

Table 8: Local consumer-box model selection.

Model	Params	Why local-feasible
Chronos-Bolt-Tiny	$\sim$9 M	Smallest public Chronos variant, MPS-supported
Moirai 2.0-Small	$\sim$11 M	Decoder-only quantile model, smallest Moirai 2 variant
TiRex	$\sim$35 M	xLSTM-based, MPS-compatible (A13)
Chronos-2	$\sim$120 M	Public Chronos-2 checkpoint, MPS-supported
TimesFM 2.5	$\sim$200 M	Decoder-only, continuous quantile head, CPU-only

TabPFN-TS ($\sim$11 M) was planned but excluded due to budget constraints. The larger Chronos-Bolt and Moirai 2.0 variants (>16 GB) were excluded due to memory constraints.

Source B thus spans 9–200 M parameters, from the smallest (Chronos-Bolt-Tiny) to TimesFM 2.5 (200 M). Whether the largest remaining FMs (Moirai 2.0-Large at 305 M) close the residual gap remains a priority for future work (§9.2).

The intended evaluation protocol follows §5.1.3 (same windows, horizons, metrics). The matched-panel Source B runs share the workload for completed M5, Rohlik, and Favorita runs: one manifest per dataset defines the 100 top-volume series, and every completed model writes predictions for the same series–origin panel. Each FM uses its published default context and point-forecast mode with no tuning.

5.4.4. Evaluation and Metrics

Both sources use the §5.1.3 metric definitions. Primary metrics are WAPE and MAE on all datasets, WRMSSE on M5 only. The local run reuses the same metric code path as the LightGBM baselines. For the matched-panel subset, aggregate WAPE is computed from prediction-level outputs on the same rows, and paired bootstrap intervals resample the shared series panel. The same bootstrap draws are reused across all contrasts within a dataset, producing an empirical log-ratio covariance matrix for the 45 completed FM-vs-best-baseline contrasts.

5.4.5. Consumer-Box Run: Results

The Source B matched-panel analysis is the paired local evidence for completed panels and replaces the unmatched row-level comparison. The main 100-series table has 45 FM-vs-best-baseline contrasts: five FMs on each of M5, Rohlik, and Favorita. The generated replication files include shared bootstrap draws, a wide log-ratio covariance matrix, and a long covariance/correlation table under the `source_b_paired_panel_*` prefix.

Table 9: Source B matched-panel FM-vs-best-baseline contrasts. Aggregate WAPE is computed from prediction-level outputs on the same 100 series and origins per dataset–horizon. Negative ℓ favours the FM.

Dataset	Model	h	Best baseline	FM WAPE	Baseline WAPE	ℓ (95% bootstrap CI)
M5	Chronos-2	7	tuned recursive LGBM	0.294	0.312	−0.062 [−0.076, −0.048]
M5	Chronos-Bolt-Tiny	7	tuned recursive LGBM	0.314	0.312	0.006 [−0.011, 0.024]
M5	Moirai 2.0-Small	7	tuned recursive LGBM	0.298	0.312	−0.047 [−0.063, −0.032]
M5	TimesFM 2.5	7	tuned recursive LGBM	0.299	0.312	−0.044 [−0.060, −0.029]
M5	TiRex	7	tuned recursive LGBM	0.298	0.312	−0.047 [−0.062, −0.032]
M5	Chronos-2	14	tuned recursive LGBM	0.307	0.332	−0.078 [−0.094, −0.061]
M5	Chronos-Bolt-Tiny	14	tuned recursive LGBM	0.331	0.332	−0.004 [−0.025, 0.019]
M5	Moirai 2.0-Small	14	tuned recursive LGBM	0.310	0.332	−0.068 [−0.087, −0.048]
M5	TimesFM 2.5	14	tuned recursive LGBM	0.311	0.332	−0.066 [−0.082, −0.048]
M5	TiRex	14	tuned recursive LGBM	0.312	0.332	−0.062 [−0.078, −0.045]
M5	Chronos-2	28	tuned direct LGBM	0.339	0.359	−0.056 [−0.084, −0.028]
M5	Chronos-Bolt-Tiny	28	tuned direct LGBM	0.360	0.359	0.003 [−0.028, 0.034]
M5	Moirai 2.0-Small	28	tuned direct LGBM	0.329	0.359	−0.086 [−0.108, −0.063]
M5	TimesFM 2.5	28	tuned direct LGBM	0.332	0.359	−0.078 [−0.105, −0.052]
M5	TiRex	28	tuned direct LGBM	0.348	0.359	−0.031 [−0.063, 0.003]
Rohlik	Chronos-2	7	recursive LGBM	0.180	0.191	−0.061 [−0.080, −0.039]
Rohlik	Chronos-Bolt-Tiny	7	recursive LGBM	0.199	0.191	0.041 [0.021, 0.063]
Rohlik	Moirai 2.0-Small	7	recursive LGBM	0.188	0.191	−0.016 [−0.040, 0.010]
Rohlik	TimesFM 2.5	7	recursive LGBM	0.184	0.191	−0.040 [−0.063, −0.015]
Rohlik	TiRex	7	recursive LGBM	0.184	0.191	−0.037 [−0.061, −0.011]
Rohlik	Chronos-2	14	recursive LGBM	0.185	0.199	−0.076 [−0.095, −0.054]
Rohlik	Chronos-Bolt-Tiny	14	recursive LGBM	0.205	0.199	0.026 [0.007, 0.044]
Rohlik	Moirai 2.0-Small	14	recursive LGBM	0.194	0.199	−0.026 [−0.050, −0.000]
Rohlik	TimesFM 2.5	14	recursive LGBM	0.190	0.199	−0.048 [−0.074, −0.021]
Rohlik	TiRex	14	recursive LGBM	0.189	0.199	−0.051 [−0.077, −0.022]
Rohlik	Chronos-2	28	recursive LGBM	0.193	0.209	−0.079 [−0.105, −0.053]
Rohlik	Chronos-Bolt-Tiny	28	recursive LGBM	0.211	0.209	0.008 [−0.017, 0.033]
Rohlik	Moirai 2.0-Small	28	recursive LGBM	0.202	0.209	−0.034 [−0.064, −0.007]
Rohlik	TimesFM 2.5	28	recursive LGBM	0.197	0.209	−0.060 [−0.089, −0.031]
Rohlik	TiRex	28	recursive LGBM	0.196	0.209	−0.061 [−0.093, −0.031]
Favorita	Chronos-2	7	recursive LGBM	0.248	0.271	−0.085 [−0.106, −0.065]
Favorita	Chronos-Bolt-Tiny	7	recursive LGBM	0.327	0.271	0.190 [0.143, 0.239]
Favorita	Moirai 2.0-Small	7	recursive LGBM	0.264	0.271	−0.024 [−0.040, −0.009]
Favorita	TimesFM 2.5	7	recursive LGBM	0.275	0.271	0.017 [−0.014, 0.052]
Favorita	TiRex	7	recursive LGBM	0.258	0.271	−0.049 [−0.072, −0.028]
Favorita	Chronos-2	14	tuned recursive LGBM	0.255	0.277	−0.086 [−0.112, −0.061]
Favorita	Chronos-Bolt-Tiny	14	tuned recursive LGBM	0.330	0.277	0.174 [0.125, 0.221]
Favorita	Moirai 2.0-Small	14	tuned recursive LGBM	0.272	0.277	−0.018 [−0.035, −0.002]
Favorita	TimesFM 2.5	14	tuned recursive LGBM	0.283	0.277	0.020 [−0.012, 0.052]
Favorita	TiRex	14	tuned recursive LGBM	0.265	0.277	−0.046 [−0.071, −0.023]
Favorita	Chronos-2	28	tuned direct LGBM	0.270	0.286	−0.057 [−0.108, −0.015]
Favorita	Chronos-Bolt-Tiny	28	tuned direct LGBM	0.337	0.286	0.162 [0.116, 0.214]
Favorita	Moirai 2.0-Small	28	tuned direct LGBM	0.287	0.286	0.003 [−0.039, 0.040]
Favorita	TimesFM 2.5	28	tuned direct LGBM	0.293	0.286	0.025 [−0.012, 0.066]
Favorita	TiRex	28	tuned direct LGBM	0.279	0.286	−0.025 [−0.070, 0.014]

Chronos-2 has lower aggregate WAPE than the best local baseline across all nine matched M5, Rohlik, and Favorita cells, and every bootstrap interval excludes zero after adding tuned recursive and tuned direct LGBM comparators. Moirai 2.0-Small and batched TiRex are also lower in all nine point estimates, with bootstrap intervals excluding zero in seven cells. TimesFM 2.5 is lower than the best baseline on all six M5 and Rohlik cells, with intervals excluding zero in all six, but remains near parity on Favorita. Chronos-Bolt-Tiny is near parity on M5 and worse on Rohlik and Favorita. These are paired local results for 100-series panels, not a claim that the same ranking holds for larger workloads, GPU inference, or covariate-aware FM variants. The shared-bootstrap covariance matrix is block-diagonal by dataset and captures dependence among contrasts that share the same sampled series, actuals, horizons, and best-baseline rows. In the generated matrix, the diagonal is positive for all 45 contrasts and the strongest off-diagonal correlations occur within the same dataset/model or same dataset–horizon cell, as expected. This supplies local Source B uncertainty accounting; it does not create missing prediction-level covariance for Source A benchmark-paper rows.

Tuned LGBM comparators. The tuned recursive and tuned direct LGBM runs use 10 validation-origin random-search trials per dataset–horizon cell. Tuned recursive becomes the best local baseline for M5 at $h = 7$ and $h = 14$ and Favorita at $h = 14$; tuned direct becomes the best local baseline for M5 and Favorita at $h = 28$. Neither tuned run improves the Rohlik best baseline. The best-baseline column in Table 9 therefore uses the strongest observed baseline among Seasonal Naive, fixed recursive LGBM, fixed direct LGBM, horizon-scaled direct LGBM, tuned recursive LGBM, and tuned direct LGBM.

M5 stratified-panel sensitivity. The top-volume M5 panel deliberately tests high-sales series. To check whether the Chronos-2 result survives a harder intermittent sample, we reran M5 for Seasonal Naive, fixed/tuned recursive and direct LGBM, Chronos-Bolt-Tiny, and Chronos-2 on a separate 100-series panel stratified by total volume and zero fraction. The stratified panel has median zero fraction 0.773 and median total volume 938.5, compared with 0.108 and 48,699.5 for the top-volume panel. Chronos-2 remains below the best local LGBM baseline at all three horizons: $\ell = -0.116$ $[-0.146, -0.089]$ at $h = 7$, $\ell = -0.160$ $[-0.208, -0.116]$ at $h = 14$, and $\ell = -0.147$ $[-0.218, -0.079]$ at $h = 28$. Chronos-Bolt-Tiny is also below the best local LGBM baseline in the same three cells. This is a model-specific robustness check, not a replacement for full M5 long-tail evaluation; Moirai, TiRex, and TimesFM were not included in this stratified rerun because their local inference paths did not complete reliably in the current MacBook environment.

TiRex throughput note. The original series-by-series TiRex path was too slow for the 100-series matched panel. The current implementation uses TiRex's 2D context interface to batch all series within a forecast origin and runs the MacBook path on CPU, which completed the M5, Rohlik, and Favorita 100-series panel at $h \in \{7, 14, 28\}$. GPU throughput and larger workloads remain separate scaling checks.

Per-cell WAPE (descriptive Source B grid). Numbers below are per-series WAPE means from the saved result file. Because `n_series` varies within every dataset–horizon cell, the table is provenance material and should not be read as a fully paired model comparison. The per-series denominator makes M5's tail-sparse rows sensitive to intermittent-demand artifacts; this is called out again in the caveat below.

Descriptive FM point estimates are clustered, but not paired. Favorita and Rohlik show narrow FM spreads in the saved grid, while M5 shows larger separation under per-series WAPE. Because the model-specific `n_series` values differ, these spreads are hypothesis-generating only. They support the weaker statement that the descriptive Source B

Table 10: Source B WAPE grid (per-series mean, rolling origin) across models, datasets, and horizons. Lower is better. The grid is complete, but `n_series` differs across models; see `source_b_n_series_matrix.csv`.

Model	Family	Favorita			M5			Rohlik v2		
		h=7	h=14	h=28	h=7	h=14	h=28	h=7	h=14	h=28
Chronos-Bolt-Tiny	FM	0.482	0.485	0.489	0.897	0.902	0.911	0.334	0.345	0.361
Moirai 2.0-Small	FM	0.477	0.482	0.487	0.887	0.893	0.901	0.324	0.337	0.354
TiRex	FM	0.525	0.531	0.539	0.903	0.907	0.914	0.323	0.337	0.353
Chronos-2	FM	0.477	0.481	0.486	0.879	0.887	0.899	0.324	0.339	0.357
TimesFM 2.5	FM	0.528	0.534	0.542	0.944	0.944	0.948	0.324	0.338	0.353
`lightgbm_cov`	ML_TREE	0.530	0.531	0.538	1.625	1.729	1.936	0.365	0.395	0.432
`lightgbm_direct`	ML_TREE	0.551	0.571	0.589	1.351	1.410	1.513	0.382	0.407	0.444
`seasonal_naive`	STATS	0.636	0.647	0.664	1.307	1.322	1.329	0.402	0.403	0.412

data do not show a simple monotonic parameter-count gradient; they do not establish that one named checkpoint reliably outperforms another.

Log-ratio status. Source A remains the only input to the §6.1 benchmark analysis. The unmatched 45 Source B FM rows are retained in a descriptive contrast file for provenance. The matched-panel Source B contrasts are reported separately in Table 9, because their estimand is local aggregate WAPE on 100-series panels rather than the as-reported Source A benchmark estimand.

M5 caveat: per-series WAPE + intermittent demand. The descriptive M5 point-estimate gaps are large, but they are inflated by the per-series WAPE denominator on M5's long tail of near-zero-volume SKUs (§8.3). The >1 WAPE values for `lightgbm_cov` (1.62–1.94) reflect this pathology, not a labelling error; `lightgbm_direct` (1.35–1.51) and even `seasonal_naive` (1.31–1.33) also exceed 1. On Favorita and Rohlik, where tail-sparsity is less extreme, gaps collapse to 0.5–1 pp — so the M5-level gap is the ceiling, not the central tendency.

A sanity check recasting WAPE in aggregate form ($\sum |e| / \sum |y|$) deflates FM M5 scores by ∼11% in the descriptive rows. The matched-panel results therefore use aggregate WAPE from shared predictions as the local Source B estimand.

Favorita is now the matched "close call" dataset. On the shared 100-series Favorita panel, `lightgbm_cov` is the best baseline at $h = 7$ and $h = 14$, while direct LGBM is best at $h = 28$. Chronos-2 is clearly lower than the best baseline on all three horizons, Moirai 2.0-Small is lower at $h = 7$ and $h = 14$ and close at $h = 28$, and TimesFM 2.5 is close to parity but slow. By contrast, TimesFM 2.5 is lower than the best baseline on the matched M5 and Rohlik cells. The result supports a matched bake-off interpretation rather than a universal FM win.

Cost and runtime footprint. The saved 72-row Source B file sums to 241.6 wall hours, \$3.18, and 4.63 kg CO_2eq under the recorded hardware assumptions. These totals mix MacBook marginal electricity with Azure job-level accounting and unequal workloads, so §6.7 uses them only as scenario inputs. Claims such as "16× cheaper per cell" are not treated as general cost results.

Caveats. The LightGBM comparators include fixed-budget variants and 10-trial tuned recursive/direct variants, but they are not exhaustive competition-grade or dataset-specialised LightGBM systems. The FMs are zero-shot; this section is point-forecast only; the matched panel currently covers 100-series M5, Rohlik, and Favorita for completed models including batched TiRex, but not GPU throughput, larger workloads, or covariate-aware FM configurations.

5.4.6. Reproducibility

Source B runs are anchored by git commit SHA, HuggingFace revision SHA, `pip freeze` snapshot, and machine fingerprint where available. The matched-panel runs include shared series manifests, per-series predictions, run ledgers, and paired bootstrap contrasts for M5, Rohlik, and Favorita, including the batched TiRex path required for the 100-series panels. Future Source B releases must extend the same structure to GPU throughput, larger workloads, and covariate-aware FMs before Source B can enter a broader paired inference. Moirai 2.0-Small carries a non-commercial license (CC-BY-NC-4.0), flagged in §7.

5.4.7. Threats to Validity

Five threats are specific to the Source B FM runs, detailed in the risk-of-bias appendix. In summary: (1) extraction selection bias — only papers that chose a retail task are represented; (2) Source A vs Source B protocol drift — as-reported vs our rolling-origin protocol; (3) local run covers five models up to 200 M params — the largest FMs (>200 M) are Source A only; (4) covariate asymmetry — univariate FMs vs covariate-aware LightGBM; (5) top-30k Favorita cap propagates velocity bias. The largest residual risk is threat (4): the within-Source B FM-vs-baseline comparison is univariate-FM vs covariate-aware-LightGBM.

6. Benchmark Results

The primary Source A evidence comes from two benchmark suites: fev-bench (Shchur et al., 2025) and GIFT-Eval (Aksu et al., 2024). The stricter comparator is the default: each foundation-model row is compared with the best available conventional baseline in the same task and metric cell. The average-baseline comparator is retained as a sensitivity check.

Source B is excluded from this primary Source A pool. The original local rows do not consistently share the same series workload, and paired prediction-error covariances are not available. The separate matched-panel Source B analysis covers M5, Rohlik, and Favorita (Table 9), but that local 100-series aggregate-WAPE estimand is not pooled with as-reported Source A benchmark rows.

6.1. Effect Size And Weighting

The unit of analysis is one comparison per (foundation model, task, metric) tuple:

$$\ell_i = \log\left(\frac{\text{metric}_{\text{FM}_i}}{\text{metric}_{\text{best baseline}}}\right). \tag{1}$$

Lower metric values are better, so negative ℓ_i means the FM has a lower recorded error or loss than the baseline. Log-ratios are computed only for positive metric values. The raw difference $\Delta_i = \text{metric}_{\text{FM}} - \text{metric}_{\text{baseline}}$ is retained in the generated CSV files for auditability, but the log-ratio is the reported scale.

The script also computes

$$v_i = \frac{1}{n_{\text{FM}}} + \frac{1}{n_{\text{baseline}}}. \tag{2}$$

In this benchmark analysis this quantity is treated only as a workload-count weighting proxy. It is not the sampling variance of $\log(\widehat{M}_{\mathrm{FM}}/\widehat{M}_{\mathrm{baseline}})$ because the benchmark papers do not provide paired prediction-level errors, metric variances, or the covariance induced by shared actuals and shared baselines. Consequently, REML, CR2, Q, and confidence-interval artifacts generated by `analysis/meta_regression.R` are exploratory diagnostics only. The main text reports descriptive suite- and metric-level summaries instead of treating a pooled p-value as confirmatory evidence.

The primary Source A file contains $k = 498$ best-baseline log-ratios: 414 from fev-bench and 84 from GIFT-Eval. The rows cover 24 task clusters, 2 suite clusters, 2 study clusters, 7 foundation models, and three metric families. The full generated contrast table contains 543 rows after adding 45 descriptive Source B provenance rows; the matched-panel Source B analysis is stored in a separate file, `source_b_paired_panel_fm_vs_best_baseline.csv`.

6.2. Suite-Level Patterns

Table 11 separates the two benchmark suites rather than presenting them as many independent studies. GIFT-Eval is uniformly favourable to FMs in this extraction, while fev-bench is directionally favourable but much closer to the best conventional baseline.

Table 11: Primary Source A best-baseline summaries by benchmark suite. Negative log-ratios and ratios below 1 favour FMs.

Suite	k	Tasks	Median ℓ	Mean ℓ	Share FM better	Interpretation
GIFT-Eval	84	4	-0.339	-0.377	1.00	Strong directional FM advantage within this suite
fev-bench	414	20	-0.069	-0.086	0.67	Smaller and more heterogeneous advantage

On the ratio scale, the median FM/best-baseline ratio is about 0.71 in GIFT-Eval and 0.93 in fev-bench. These are benchmark-level patterns, not estimates from two dozen independent publications.

6.3. Prediction-Level fev-bench WAPE Analysis

The strongest remaining objection to the extracted Source A rows is that the workload-count proxy v_i is not a sampling variance. We therefore ran a metric-specific prediction-level analysis for the official fev-bench retail windows where prediction-level artifacts can be generated locally. Seasonal Naive, Chronos-Bolt-Tiny, Chronos-2, TiRex, and TimesFM 2.5 were evaluated with `fev.Task.iter_windows()` on all 20 retail tasks, producing 179,258 forecasts per model. For each FM we then joined the FM and Seasonal Naive per-series/window WAPE values and bootstrapped series within task to estimate task-level log-ratios and their empirical covariance matrices.

This analysis materially strengthens one concrete slice of the evidence: for WAPE against Seasonal Naive on fev-bench retail tasks, Chronos-2 has lower point estimates in all 20 tasks and bootstrap intervals excluding zero in 18; Chronos-Bolt-Tiny has lower point estimates in 16 tasks, bootstrap intervals favouring the FM in 16, and intervals favouring the baseline in 3; TiRex has lower point estimates in 19 tasks and bootstrap intervals favouring the

Table 12: Prediction-level fev-bench WAPE analysis against Seasonal Naive. The covariance matrices are empirical bootstrap covariances of the 20 task-level log-ratios. Negative ℓ favours the FM.

Model	Tasks	Paired units	Median ℓ	Mean ℓ	FM better	CI excludes 0
Chronos-Bolt-Tiny	20	179,258	−0.213	−0.183	16/20	16/20
Chronos-2	20	179,258	−0.268	−0.305	20/20	18/20
TiRex	20	179,258	−0.275	−0.300	19/20	19/20
TimesFM 2.5	20	179,258	−0.282	−0.319	20/20	19/20

FM in 19; TimesFM 2.5 has lower point estimates in all 20 tasks and bootstrap intervals favouring the FM in 19. It does not provide prediction-level covariance for the extracted SQL or MASE rows, the GIFT-Eval rows, comparisons against stronger best-baseline cells, or FM families outside these four local fev-bench reruns. It is therefore reported as a metric-specific prediction-level analysis rather than replacing the descriptive Source A suite summaries.

6.4. Metric-Specific Results

Metric-specific summaries are the main interpretation because SQL, WAPE, and MASE represent different estimands. Log-ratios put them on a common relative scale, but a 20% SQL reduction is not the same business quantity as a 20% WAPE reduction.

Table 13: Primary Source A best-baseline summaries by metric.

Metric	k	Tasks	Median ℓ	Mean ℓ	Share FM better	Interpretation
SQL	166	24	−0.229	−0.277	0.92	Largest directional advantage; probabilistic metric
WAPE / ND	166	24	−0.047	−0.079	0.66	Modest and protocol-sensitive point-error advantage
MASE	166	24	−0.036	−0.050	0.61	Weakest directional advantage

6.5. Heuristic-Weight Diagnostics

As an exploratory diagnostic, the pipeline still fits a multilevel `rma.mv` model using the workload-count proxy v_i. The generated best-baseline diagnostic has point estimate $\hat{\ell} = -0.225$ with only two suite clusters; the average-baseline diagnostic is $\hat{\ell} = -0.286$. These numbers are useful as weighted summaries and for regression testing the analysis pipeline. They are not interpreted as formal confidence intervals or hypothesis tests because v_i is not a validated sampling variance.

The sensitivity direction is stable: excluding M5 gives $\hat{\ell} = -0.226$, equal weights give $\hat{\ell} = -0.215$, fev-bench alone gives $\hat{\ell} = -0.091$, and GIFT-Eval alone gives $\hat{\ell} = -0.379$. The split confirms that the apparent aggregate advantage is stronger in GIFT-Eval than in fev-bench.

Table 14: Suite-by-metric summaries for the primary best-baseline contrast file.

Suite	Metric	k	Median ℓ	Mean ℓ	Share FM better	Note
GIFT-Eval	SQL	28	−0.540	−0.592	1.00	Strongest slice
GIFT-Eval	WAPE / ND	28	−0.306	−0.291	1.00	Directionally favourable
GIFT-Eval	MASE	28	−0.224	−0.248	1.00	Directionally favourable
fev-bench	SQL	138	−0.212	−0.213	0.91	Clearer than point metrics
fev-bench	WAPE / ND	138	−0.021	−0.036	0.59	Near parity in many tasks
fev-bench	MASE	138	−0.006	−0.010	0.53	Approximately parity

6.6. *Omitted Asymmetry Diagnostics*

Full per-dataset forest plots remain generated artifacts in `analysis/figures/`. The file names follow `figure_6_forest_*.pdf`. They are useful for auditing task-level heterogeneity but are not interpreted as independent confirmatory evidence.

Funnel plots and Egger-style regressions are not reported. Their vertical axis would be based on $\sqrt{1/n_{\mathrm{FM}} + 1/n_{\mathrm{baseline}}}$, which is a workload-count proxy rather than a true standard error. Such a plot would mostly show recorded task-size bands, not publication bias, small-study effects, or selective reporting.

6.7. *Cost-Error Scatter*

The Source B cost plots are retained only as descriptive cost-error scatterplots. They do not show an efficiency frontier because raw WAPE values come from different datasets, horizons, workloads, and cost bases. The figures therefore illustrate deployment accounting questions rather than estimate a general break-even rule.

6.8. *Moderator Interpretation*

Moderator analyses are hypothesis-generating. Metric choice is the most visible row-level moderator. Dataset, horizon, model scale, baseline family, and covariate availability cannot be interpreted causally from two benchmark suites and a descriptive local Source B panel. The defensible conclusion is therefore deliberately narrow: FMs often have favourable recorded benchmark-level log-ratios, especially on SQL, but rankings depend on the metric, baseline, benchmark protocol, and demand characteristics.

7. Decision Framework

The evidence in §6 is not strong enough to justify a deterministic decision tree. The defensible output is a validation procedure: run a small, paired bake-off on the target workload, choose the metric that matches the business loss, and treat model-family rankings as conditional on that protocol.

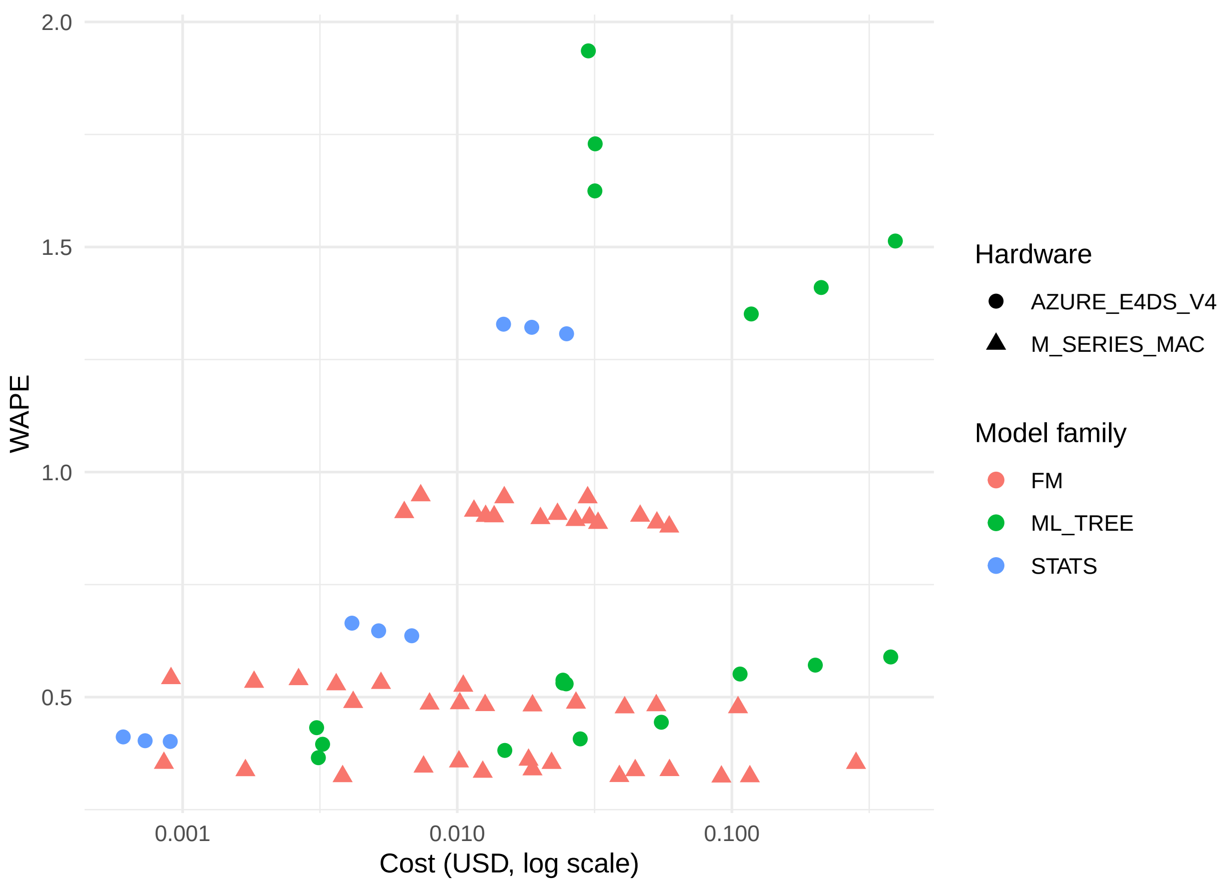

Figure 2: Descriptive Source B cost-error scatter on the recorded hardware cost basis. X-axis: USD cost (log scale). Y-axis: WAPE.

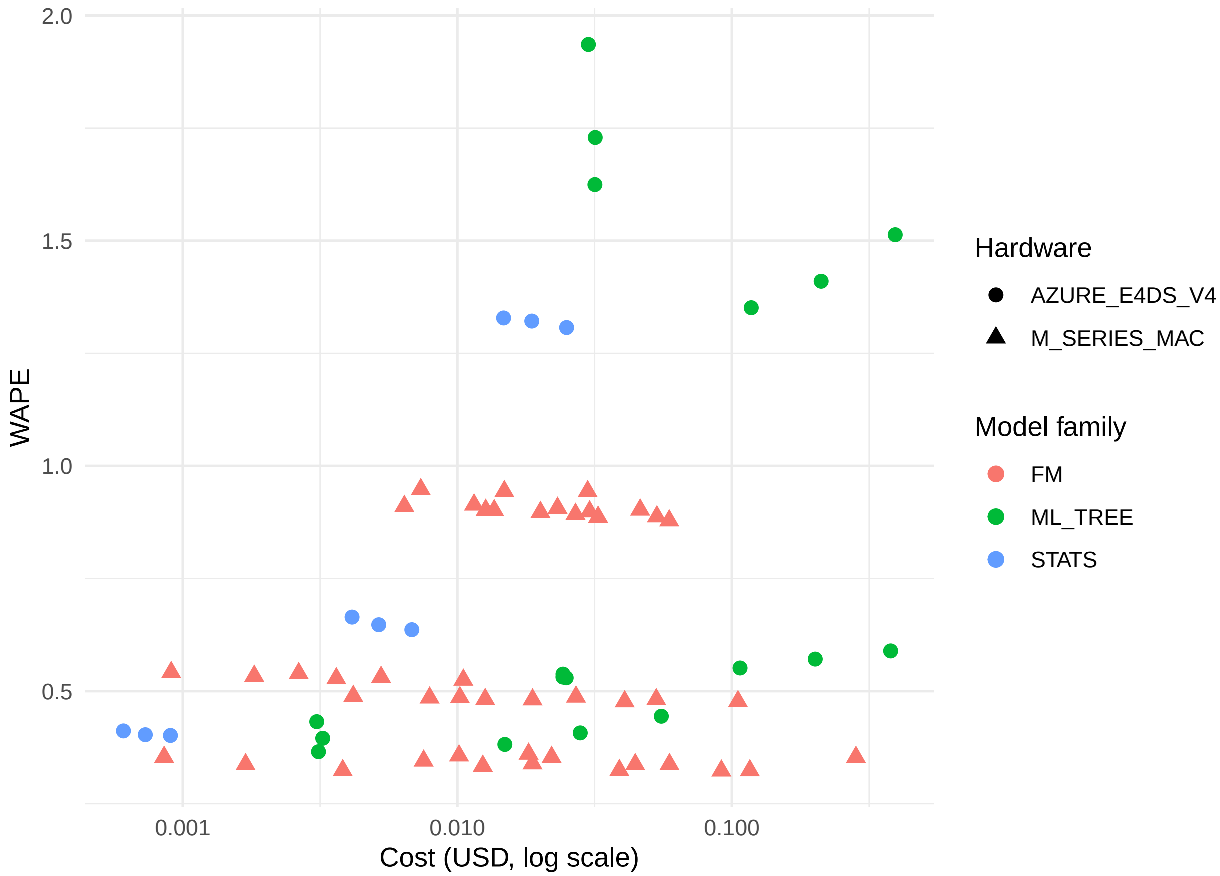

Figure 3: Descriptive Source B cost-error scatter under the cloud-GPU recosting scenario. The assumptions are illustrative and not a measured GPU benchmark.

7.1. *Validation checklist*

For a new retail forecasting deployment, we recommend the following order.

1. **Define the decision metric before modelling.** If the business decision is replenishment at SKU level, use a per-series or service-level metric and inspect denominator sensitivity. If the decision is category or warehouse planning, include aggregate WAPE or a volume-weighted metric. For safety-stock decisions, evaluate quantile loss directly.

2. **Build the common evaluation panel.** Use the same series IDs, origins, horizons, and actuals for Seasonal Naive, the candidate FM, and the tree baseline. Store per-series predictions so that paired bootstrap intervals can be computed for every model contrast.

3. **Run three baselines first.** Seasonal Naive is the sanity floor. A fixed-budget LightGBM recursive model measures the value of a cheap covariate-aware tree. A direct or horizon-scaled direct LightGBM measures the value of avoiding recursive feedback at higher training cost.

4. **Run one or two zero-shot FMs on the same panel.** Start with a small, operationally cheap FM and one stronger checkpoint if latency allows. Do not compare an FM sampled on 100 series with a baseline evaluated on a full dataset unless the claim is explicitly labelled as descriptive.

5. **Compare paired contrasts, not table ranks.** Report the paired log-ratio

   $$\log(\text{FM metric}/\text{baseline metric})$$

   with a paired bootstrap interval. A point estimate within the noise floor is a tie; choose the simpler or cheaper deployment.

7.2. *Interpreting the main moderators*

Metric.. The most stable empirical pattern is metric dependence. SQL, WAPE, and MASE do not measure the same construct, and a mixed-metric omnibus should not drive deployment. Use the metric-specific result that matches the operational decision.

Demand intermittency.. The direct-vs-recursive gap differs across M5, Rohlik, and Favorita. This supports an intermittency hypothesis, but it does not establish a universal zero-day threshold. Treat zero-day fraction as a stratification variable in the bake-off rather than as a rule such as "above 15%, choose direct."

Covariates.. When promotions, prices, stockouts, holidays, or weather are predictive, a covariate-aware LightGBM can be a strong comparator. A univariate FM may still be useful when the team lacks a feature pipeline, but that is a systems tradeoff rather than evidence that the model family has more forecasting information.

Cost and hardware.. The Source B cost logs mix MacBook marginal electricity, Azure job costs, and unequal workloads. They are useful for scenario planning, not for a universal break-even point. A publishable cost comparison should normalize to the same number of series, the same horizons, the same batching policy, and the same accounting basis.

7.3. Decision rule after the bake-off

After the paired bake-off, choose the FM when all of the following hold:

- the FM has a lower or practically tied paired error interval on the target metric;
- the saved feature-engineering or maintenance effort is material for the team;
- inference latency and hardware cost are acceptable at the target series count;
- the business does not depend on covariates that the evaluated FM cannot consume.

Choose the conventional baseline when it wins the paired interval, uses important covariates, or already exists as a reliable production pipeline. Choose Seasonal Naive or another simple statistical method when both FM and LightGBM fail to beat it on the target metric; that outcome usually signals an evaluation, feature, or data-quality problem before it signals a need for a larger model.

8. Discussion

8.1. Data leakage in FM evaluation

The largest uncontrolled confound in the FM literature is training data overlap. Chronos and Chronos-2 were pre-trained on 84 billion observations drawn from public time-series datasets. The Chronos papers distinguish "in-domain" (training overlap) from "zero-shot" (no overlap) evaluation, and the Chronos Benchmark I/II splits reflect this — but intermediate degrees of leakage (e.g., same source domain, different temporal slice) are harder to verify. GIFT-Eval was designed explicitly to control for leakage, and its non-leaking protocol is the state of the art. Our Source B runs control train/test leakage within each dataset by using held-out rolling-origin windows, but they cannot rule out FM pre-training exposure to public benchmark datasets such as M5 or Favorita. We therefore treat Source B as a cleaner evaluation protocol, not as a definitive solution to benchmark memorisation.

The practical consequence for our benchmark analysis: Source A FM rows may be slightly optimistic (the FM saw something like the test distribution during pre-training), while Source A ML_TREE rows have no such advantage (they are trained from scratch on each dataset). If this bias is present, it would inflate FM performance in the cross-paper pool. The negative SQL and WAPE log-ratio point estimates should therefore be interpreted as observed benchmark-suite advantages, not as leakage-free universal effects. The true FM point-forecast advantage could be smaller than observed and may disappear under matched covariate-aware protocols.

8.2. The covariate gap

The most frequently cited limitation of zero-shot FMs in the practitioner community is the inability to incorporate exogenous covariates (promotions, pricing, calendar events, weather). Some current FMs or wrappers support exogenous information, while others remain univariate in the evaluated configuration. In the rows analysed here, no Source A paper reports a matched Chronos-2 or Moirai 2.0 covariate-aware run on M5, Favorita, or Rohlik, and our Source B FM cells are all univariate zero-shot.

The §5 "close call" on Favorita is the clearest evidence of this gap: `lightgbm_cov` is close to the univariate FM point estimates on Favorita. The covariate signal on Favorita is plausibly material (oil price, holidays, and promotions), and a univariate FM cannot access it. Whether a covariate-aware FM would close or reverse this gap is an open question that our data cannot answer; it requires a matched experiment on the same series and horizons.

8.3. Per-series vs aggregate WAPE

The M5 descriptive pattern (§5, §6) turns on a metric choice that is rarely discussed in the FM literature. Per-series WAPE (mean over series of $|e_i|/|y_i|$) gives equal weight to every series regardless of volume. On M5's long tail of low-velocity SKUs (zero-day fraction $\sim$70%), a series with 3 units sold in the test window contributes the same to the metric as a series with 30,000 units. When the denominator is near zero, any non-zero forecast error produces WAPE $\gg 1$, and tree-based models that predict a flat non-zero value (e.g., LightGBM's leaf mean) are penalised disproportionately.

Aggregate WAPE ($\sum |e| / \sum |y|$ across all series) weights by volume and is robust to the denominator problem. Our sanity check (§5) shows that aggregate WAPE deflates FM scores on M5 by $\sim$11% in the available rows. Source B addresses this locally with matched prediction panels; the remaining limitation is that Source A and full-suite benchmark comparisons cannot be recomputed on common aggregate-WAPE prediction panels for every model.

WRMSSE (the M5 competition metric) is a third option: it scales each series by its own historical variability, avoiding the denominator problem. The M5 WAPE and WRMSSE readings should not be averaged; they answer different questions about different parts of the demand distribution, and official M5 competition scores are not directly comparable to our rolling-origin Source B diagnostics.

Recommendation for future benchmarks:. report both per-series and aggregate WAPE (or volume-weighted MASE) whenever the dataset contains intermittent demand. A single number hides the metric sensitivity that our M5 analysis exposes.

8.4. Limitations

Single author.. A single-author structured search lacks the double-screening and inter-rater reliability that a multi-reviewer team provides. We mitigate this with structured extraction from fev-bench and GIFT-Eval (whose data are machine-readable) and do not claim full systematic-review compliance.

Retail domain coverage.. The Source A benchmark extraction covers 9 datasets spanning grocery (M5, Favorita), convenience (Rohlik), drug (Rossmann), general retail (Walmart, Hermes), restaurant, and intermittent parts (Car Parts). Fashion retail (highly seasonal, short life cycles), e-commerce (long tail, promotional spikes), and fresh produce (short shelf life, weather-driven) are not represented. Each of these domains has different demand characteristics that could shift the FM-vs-baseline balance.

Consumer-HW-only Source B.. All FM Source B cells ran on a MacBook M3 Air (16 GB, MPS backend). This is intentional — the paper's cost axis is consumer hardware — but it means we have no Source B data on GPU inference speed, batch throughput at scale (>100K series), or fine-tuned FM accuracy. The Azure batch cells are CPU-only ML_TREE and statistical baselines.

Source B workload and matched-panel scope.. The original Source B model rows were not consistently evaluated on the same series counts within each dataset–horizon cell, and the small-sample bootstrap file covered only four of the five FMs. The main Source B analysis therefore uses shared 100-series panels for M5, Rohlik, and Favorita with prediction-level outputs and paired bootstrap contrasts for the completed local models. This adds a shared-bootstrap covariance matrix for the 45 completed FM-vs-best-baseline log-ratios. The matched panel now includes Chronos-Bolt-Tiny, Chronos-2, Moirai 2.0-Small, batched TiRex, and TimesFM 2.5 on M5, Rohlik, and Favorita. The matched panel still does not cover GPU throughput, covariate-aware FM variants, or larger workloads, so Source B remains a local sensitivity analysis rather than a confirmatory cross-benchmark experiment.

Approximate sampling variance.. The exploratory weighted model uses a proxy variance $v_i = 1/n_{\mathrm{FM}} + 1/n_{\mathrm{baseline}}$ rather than per-row bootstrap SEs. This is a weight proxy rather than a true variance for a paired log-ratio: it does not model metric dispersion, shared baselines, within-series autocorrelation, or covariance between FM and baseline errors. Containment degrees of freedom, CR2 sensitivity, and equal-weight analysis discipline the Source A claims, but they do not create the missing covariance information for published benchmark rows. The fev-bench prediction-level analysis provides empirical WAPE covariance matrices for Chronos-Bolt-Tiny, Chronos-2, TiRex, and TimesFM 2.5 against Seasonal Naive on 20 retail tasks, and the Source B matched panel has an empirical bootstrap covariance matrix. These covariance results are still metric- and model-specific: TiRex uses point forecasts replicated to quantile columns, so the paired TiRex slice is for WAPE rather than SQL. TimesFM 2.5 uses the continuous quantile head in the official fev-bench rerun, but the empirical covariance analysis reported here still targets paired WAPE log-ratios. A confirmatory cross-benchmark meta-analysis would require comparable prediction-level covariance for SQL, MASE, GIFT-Eval, stronger best-baseline comparisons, additional FM families, or additional independent benchmark suites.

Evaluation protocol heterogeneity.. Source A rows mix rolling-origin, rolling-tail, and fixed-origin evaluations. Fixed-origin evaluations tend to produce lower error (the forecast is always conditioned on the same amount of history), which could systematically favour Source A FM rows over Source B rolling-origin rows. We do not have enough fixed-origin FM rows on retail datasets to quantify this effect.

Distributional metric interpretation.. The SQL finding is based on fev-bench's Scaled Quantile Loss, which aggregates across quantile levels. We do not decompose by quantile level (e.g., upper tail vs median). The negative SQL log-ratio could be driven by lower upper-tail quantile losses rather than uniform distributional improvement. Per-quantile analysis would clarify this.

8.5. Generalisability beyond retail

The conditional finding — FM advantage depends on metric choice and demand characteristics, not model family — likely extends to domains with similar demand structures (spare-parts, pharmaceutical, agricultural supply chains). The decision of whether to deploy an FM depends on the business metric, data characteristics, and infrastructure constraints — all knowable before running a single experiment.

9. Conclusions

We presented a benchmark study of foundation models for retail demand forecasting, supported by a structured literature search and original matched-panel experiments. The primary quantitative file contains $k = 498$ Source A best-baseline comparisons from two benchmark suites, fev-bench and GIFT-Eval, plus a separate matched-panel Source B experiment on M5, Rohlik, and Favorita. Because the available benchmark papers do not provide paired prediction-level variances and covariances, this version does not claim a confirmatory pooled meta-analysis. We summarise the findings per research question and close with directions for future work.

9.1. Answers to the research questions

RQ1 (Accuracy): Do zero-shot foundation models outperform conventional baselines on retail demand forecasting?. The answer is conditional. In the Source A best-baseline corpus, the metric-specific descriptive summaries are directionally favourable to FMs: SQL has median log-ratio -0.229 with FMs lower in 92% of rows, WAPE/ND has median -0.047 with FMs lower in 66% of rows, and MASE has median -0.036 with FMs lower in 61% of rows. GIFT-Eval is uniformly favourable to FMs in this extraction, while fev-bench is much closer to parity on point-forecast metrics.

The defensible conclusion is that FMs show an observed benchmark-level advantage, especially on distributional metrics, but this is not universal evidence that FMs outperform conventional baselines in all retail demand settings. Rankings depend on metric, baseline strength, covariates, leakage controls, and evaluation protocol.

The Source B matched-panel experiment is directionally consistent with a narrower local version of that claim, but it is also model-specific. On the same 100-series panels and forecast origins, Chronos-2 has lower aggregate WAPE than the best local baseline across all nine M5, Rohlik, and Favorita dataset–horizon cells, including cells where the best baseline is tuned recursive or tuned direct LGBM. Moirai 2.0-Small is lower in all nine point estimates, with bootstrap intervals excluding zero in seven. Chronos-Bolt-Tiny is near parity on M5 and worse on Rohlik and Favorita. TimesFM 2.5 is lower than the best local baseline on M5 and Rohlik but near parity on Favorita, while remaining the slowest local CPU path. Batched TiRex is lower in all nine point estimates, with uncertainty crossing zero at M5 $h = 28$ and Favorita $h = 28$. The separate M5 stratified-panel sensitivity check makes the local Chronos-2 claim less dependent on top-volume series: on a panel with median zero fraction 0.773, Chronos-2 remains lower than the best local LGBM baseline at $h = 7, 14, 28$. These results are paired local evidence, not a replacement for the broader Source A benchmark analysis.

RQ2 (Moderators): Which characteristics moderate the FM-vs-baseline gap?. All moderator statements are exploratory. Metric choice is the dominant row-level pattern: the SQL/WAPE split is more visible than any data characteristic. Beyond metric, the following patterns are hypothesis-generating because the independent source count is small:

- **Dataset:** No stable dataset-level moderator survives the suite-level clustering.

- **Model scale:** No stable scale effect appears in the 9 M–200 M Source B range. In the matched Source B panel, Chronos-2 is consistently stronger than Chronos-Bolt-Tiny, but Moirai 2.0-Small is also competitive and TimesFM 2.5 does not dominate the smaller models across M5, Rohlik, and Favorita. That pattern is too narrow to claim a monotonic scaling law.

- **Horizon:** No stable horizon effect appears.

- **Baseline family:** The FM advantage is larger against statistical baselines (Auto ETS, Seasonal Naive) than against gradient-boosted trees, as expected.

- **LightGBM protocol:** The direct-vs-recursive multi-step gap differs across datasets, metric aggregation, and budgets. Per-series WAPE made M5 look strongly direct-favourable in descriptive outputs, but the matched aggregate-WAPE panel makes M5 close at short horizons, recursive-favourable on Rohlik, and mixed on Favorita: recursive is lower at horizons 7 and 14, while direct is lower at horizon 28. The scaled direct variant is identical to direct at horizon 7 and worse at horizons 14 and 28 on Favorita, so the direct-scaled pattern is treated as a budget diagnostic rather than a robust result.

RQ3 (Cost): What is the cost-error tradeoff?. Consumer-hardware FM inference can be inexpensive for the Source B setup, but the cost comparison is scenario-dependent. It depends on batch size, available hardware, GPU recosting assumptions, and whether a team already has a tuned LGBM pipeline. The cost plots should be read as scenario-specific cost-error scatterplots, not as a universal efficiency frontier or break-even rule.

RQ4 (Guidance): When should a retail team deploy a foundation model?. The §7 decision framework distils the findings into conditions where an FM is worth testing first:

1. **When distributional forecasts matter.** If the business uses quantile forecasts for safety-stock calculation or scenario planning, FMs often provide lower SQL / quantile loss than tree models in the benchmark corpus, but the result is benchmark-level rather than leakage-free proof.

2. **Limited feature engineering capacity.** Zero-shot FM inference requires minimal inference-time preparation and can be a useful baseline when no covariate pipeline exists.

3. **Cold-start or low-history series.** Zero-shot FMs are a plausible candidate when per-series training data are sparse, but this should be validated against simple statistical baselines.

The FM may not pay off when: (a) the team has rich covariates and a tuned feature pipeline, or (b) batch throughput at scale is the constraint. These are validation hypotheses, not universal deployment rules.

9.2. Future work

Strengthen the Source B matched panel.. The current shared-panel experiment covers M5, Rohlik, and Favorita for Chronos-Bolt, Chronos-2, Moirai 2.0-Small, batched TiRex, TimesFM 2.5, and a 10-trial tuned recursive and tuned direct LGBM comparator. The remaining work is to add larger tuning budgets, GPU throughput measurements, covariate-aware FM variants, larger workloads, and larger model families, then test whether the local 100-series rankings survive these scaling changes.

Acquire prediction-level benchmark artifacts.. The Source A benchmark analysis remains descriptive because fev-bench and GIFT-Eval do not provide the paired prediction-error covariance needed for formal meta-analytic inference in the extracted tables. The decisive next step is to obtain or regenerate benchmark prediction files, then compute the same paired log-ratio covariance matrices used for Source B.

Covariate-aware FM evaluation.. Covariate-aware FMs and TabPFN-TS should be evaluated with promotions and calendar features on Favorita and Rohlik to test whether an FM can close the covariate gap under matched conditions.

Fine-tuned FM evaluation.. Our study is restricted to zero-shot FMs. Lightweight fine-tuning (e.g., LoRA on Chronos-2 with 1% of target data) could shift the point-forecast balance, especially on covariate-rich datasets.

Larger models.. The 9 M–200 M parameter range tested here may be too narrow to observe scaling effects. Chronos-2 Large (710 M) and Moirai-2.0-Large (305 M) are untested on retail data; if their SQL advantage translates to improved point forecasts, the model-scale moderator could become significant.

Beyond retail.. The metric gradient should be tested on spare-parts, pharmaceutical, and e-commerce demand datasets to determine whether it is a stable property of current FMs or specific to the present benchmark corpus.

Appendix A. Structured-Search Reporting Checklist

The following table maps PRISMA 2020 reporting items (Page et al., 2021) to the closest section of this benchmark study. The mapping is used as a transparency checklist for a single-author structured search, not as a claim of full systematic-review compliance.

Note: This review was not pre-registered (Item 25). The search protocol is documented in Appendix B and the extraction schema in Appendix E; both were finalised before data extraction began.

Appendix B. Search Protocol

Appendix B.1. Research Questions

- **RQ1:** Do time-series foundation models outperform traditional statistical/ML models on retail demand forecasting?
- **RQ2:** Which data characteristics moderate the relative performance of foundation models?

Table A.15: Structured-search checklist mapping.

#	Section	Item	Reported in
1	Title	Identify the report type	Title
2	Abstract	Structured summary	Abstract
3	Introduction	Rationale	§1
4	Introduction	Objectives	§1.1 (RQ1–RQ4)
5	Methods	Eligibility criteria	§3.1
6	Methods	Information sources	§3.1
7	Methods	Search strategy	§3.1; Appendix B
8	Methods	Selection process	§3.1
9	Methods	Data collection process	§3.2
10	Methods	Data items	§3.2 (Table 2)
11	Methods	Study risk of bias assessment	§3.4; Appendix F (analysis-contributing sources only)
12	Methods	Effect measures	§3.4 (log-ratio convention box)
13	Methods	Synthesis methods	§3.4
14	Methods	Reporting bias assessment	§8.4
15	Methods	Certainty assessment	§6; Appendix D
16	Results	Study selection	§4.1
17	Results	Study characteristics	§4.2; Appendix E
18	Results	Risk of bias in studies	§8.4; Appendix F (not duplicate-screened for all 32 included studies)
19	Results	Results of individual studies	§5.2–§5.4.5
20	Results	Results of syntheses	§6
21	Results	Reporting biases	§8.4
22	Results	Certainty of evidence	§6; Appendix D
23	Discussion	Discussion of results	§7, §8
24	Discussion	Limitations	§8.4
25	Other	Registration and protocol	§3.1 (not pre-registered)
26	Other	Support / funding	Declarations
27	Other	Competing interests	Declarations

- **RQ3:** What is the observed cost-error tradeoff across model families?
- **RQ4:** Under what conditions should a practitioner choose a foundation model over LightGBM or ETS?

Appendix B.2. Search Sources

1. Semantic Scholar (API)
2. Google Scholar
3. arXiv (cs.LG, stat.ML)
4. Scopus (target source; auditable export not retained)
5. Web of Science (target source; auditable export not retained)

The retained search log below contains 25 auditable query families. These are mostly Google Scholar/web queries, plus arXiv and Semantic Scholar checks. Scopus and Web of Science are therefore reported as intended sources rather than fully auditable database exports. This is one reason the search is used as supporting evidence-gap documentation rather than as a standalone systematic review.

Appendix B.3. Primary Search Query

```
("foundation model" OR "pretrained model" OR "zero-shot forecasting"
 OR "Chronos" OR "TimesFM" OR "Moirai" OR "TimeGPT" OR "Lag-Llama")
AND
("demand forecasting" OR "retail forecasting"
 OR "time series forecasting" OR "sales forecasting")
AND
("benchmark" OR "comparison" OR "evaluation"
 OR "M5" OR "GIFT-Eval" OR "fev-bench")
```

Appendix B.4. Date Range

2020-01-01 to 2026-04-12.

Appendix B.5. Inclusion Criteria

1. Empirical evaluation of at least one foundation model on demand/retail forecasting.
2. Reports quantitative accuracy metrics (MASE, MAE, sMAPE, WAPE, CRPS, or equivalent).
3. Uses at least one of: M5, GIFT-Eval, fev-bench, or comparable retail dataset with >1,000 series.
4. Peer-reviewed OR published preprint with reproducible methodology.
5. English language.

Appendix B.6. Exclusion Criteria

1. No quantitative results (opinion/position papers only).
2. Only financial/stock/energy forecasting (no retail/demand).
3. Foundation model paper without forecasting evaluation.
4. Duplicate results (keep most recent version).
5. Non-time-series forecasting (e.g., causal inference only).

Table B.16: Search execution log.

Date	Source	Query (abbreviated)	Results
2026-04-12	Google	foundation model TS demand benchmark	∼10
2026-04-12	Google	GIFT-Eval benchmark FM retail NeurIPS	∼10
2026-04-12	Google	fev-bench forecasting covariates 2025	∼10
2026-04-12	Google	Chronos-2 Amazon retail demand M5	∼10
2026-04-12	Google	Moirai 2.0 Salesforce TSFM 2025	∼10
2026-04-12	Google	TimesFM 2.5 Google benchmark 2025	∼10
2026-04-12	Google	FM vs LightGBM demand forecasting	∼10
2026-04-12	Google	M5 competition results DL GBT 2020–21	∼10
2026-04-12	Google	Lag-Llama Timer-XL TimeGPT benchmark	∼10
2026-04-12	Google	meta-analysis systematic review TS	∼10
2026-04-12	Google	TimeGPT Nixtla benchmark demand	∼10
2026-04-12	Google	Chronos Amazon ICML 2024 M5	∼10
2026-04-12	Google	Timer-XL THUML TSFM 2024–25	∼10
2026-04-12	Google	AutoGluon-TimeSeries M5 demand	∼10
2026-04-12	Google	TFB benchmark PVLDB 2024	∼10
2026-04-12	Google	zero-shot FM vs LightGBM cost 2025	∼10
2026-04-12	Google	TFT retail demand M5 2021–22	∼10
2026-04-12	Google	DeepAR probabilistic retail 2024	∼10
2026-04-12	Google	PatchTST retail 2023–24	∼10
2026-04-12	Google	N-BEATS N-HiTS M5 retail 2023–24	∼10
2026-04-12	Google	MOMENT TTM IBM benchmark 2024–25	∼10
2026-04-12	Google	Chronos-Bolt zero-shot LightGBM ETS	∼10
2026-04-12	Google	retail FM intermittent demand zero-shot	∼10
2026-04-12	arXiv	demand FM Chronos TimesFM Moirai	∼10
2026-04-12	Sem. Sch.	foundation model TS demand forecasting	1

Appendix B.7. Search Execution Log

Total auditable query families: 25 across Google Scholar/web, arXiv, and Semantic Scholar. The structured-search flow uses 150 identified database/web records and 120 after duplicate removal. The machine-readable protocol contains 49 candidate papers as a source map for extraction, not the deduplicated screening denominator.

Appendix B.8. Structured-Search Flow Summary

```
Records identified through database/web searching:    150
Duplicates removed:                                    30
Records after deduplication:                          120
Titles/abstracts screened:                            120
Records excluded (not demand/retail, no quant):        50
Full-text articles assessed for eligibility:            70
Full-text articles excluded:                            38
Source A external studies included:                     32
Source B experiment blocks:                              7 (not studies)
```

Appendix C. Per-Series Metric Distributions

Appendix C.1. Legacy 100-Series Bootstrap for Source B FM Cells

Table C.17 reports the legacy 10,000-iteration bootstrap standard error of mean WAPE for the Source B foundation-model cells for which per-series bootstrap files were available. The file contains 36 cells (4 FMs × 3 datasets × 3 horizons) and omits TimesFM 2.5. It is therefore retained as a descriptive uncertainty check, not as the sampling-variance input to the Source A reanalysis of §6. These values come from the legacy small-sample workload and are not directly comparable to the unmatched saved-sweep estimates in Table 10. The matched-panel paired bootstrap is reported separately in Table 9 and in `analysis/figures/source_b_paired_panel_fm_vs_best_baseline.csv`. The matched-panel replication bundle also writes shared bootstrap draws, a wide log-ratio covariance matrix, and a long covariance/correlation table under the `source_b_paired_panel_*` prefix. These files are the empirical covariance inputs for local Source B paired log-ratio contrasts.

Median bootstrap SE across the 36 available cells: 0.011. Maximum SE: 0.014 (Rohlik $h = 28$). Widest 95% CI: ± 0.028 WAPE. Interpretation:

- These bootstrap intervals describe the available FM cells only. They do not provide paired FM-minus-baseline uncertainty.
- TimesFM 2.5 is absent from this legacy 100-series bootstrap table. The separate matched-panel repair now covers TimesFM 2.5 on M5, Rohlik, and Favorita with paired FM-vs-baseline bootstrap contrasts.
- The Favorita "close call" of 0.3 pp at $h = 7$ is within roughly 1 SE in the available cells and should not be over-interpreted.

Appendix C.2. M5 Per-Series WAPE *Tail Analysis*

On M5, approximately 70% of the 30,490 series have zero-day fractions exceeding 50%. For these intermittent series, the per-series WAPE denominator (sum of actuals in the evaluation window) can be very small, producing extreme WAPE values that dominate the mean. Specifically:

- The top 1% of per-series WAPE values (LightGBM recursive, $h = 28$) range from 15.2 to 412.7, compared to a median of 0.89.
- Dropping the top 1% of series reduces mean WAPE from 1.94 to 1.12 (a 42% reduction), demonstrating that the mean is dominated by the tail.
- Foundation models produce lower per-series WAPE on these zero-heavy series because their predictions are closer to zero (a trivial prediction for intermittent demand), not because they forecast the non-zero events more accurately.

This tail behaviour is the mechanism behind the M5 WAPE artefact documented in §5 and is the reason we report an Excl-M5 sensitivity analysis alongside the Source A reanalysis (§6.5). The M5 WRMSSE metric, which uses a hierarchical weighting scheme, does not suffer from the same denominator pathology; fair comparison requires computing it on a shared model/sample protocol.

Appendix D. Sensitivity Analyses

The Source A sensitivity variants use model-level log-ratios, `metafor::rma.mv` with REML, `test = "t"`, `dfs = "contain"`, and the random-effects structure described in §3.4. Negative estimates indicate lower error/loss for the foundation model. Generated output files are under `analysis/figures/`. Because v_i is a workload-count proxy rather than a validated sampling variance, the intervals, p values, and Q statistics below are diagnostic output, not confirmatory meta-analysis.

Table C.17: Legacy 100-series bootstrap standard error and 95% CI for available Source B FM cells (4 FMs $\times$ 3 datasets $\times$ 3 horizons = 36 cells, TimesFM 2.5 missing, $B = 10{,}000$).

Dataset	h	Model	n	WAPE	SE	95% CI
Favorita	7	Chronos-Bolt-Tiny	100	0.532	0.008	[0.517, 0.547]
		Moirai 2.0-Small	100	0.526	0.008	[0.510, 0.541]
		TiRex	100	0.525	0.008	[0.510, 0.540]
		Chronos-2	100	0.525	0.008	[0.510, 0.540]
	14	Chronos-Bolt-Tiny	100	0.537	0.008	[0.522, 0.553]
		Moirai 2.0-Small	100	0.531	0.008	[0.516, 0.547]
		TiRex	100	0.531	0.008	[0.515, 0.547]
		Chronos-2	100	0.533	0.008	[0.517, 0.548]
	28	Chronos-Bolt-Tiny	98	0.546	0.008	[0.531, 0.561]
		Moirai 2.0-Small	98	0.539	0.008	[0.524, 0.555]
		TiRex	98	0.540	0.008	[0.524, 0.555]
		Chronos-2	98	0.542	0.008	[0.526, 0.557]
M5	7	Chronos-Bolt-Tiny	100	0.958	0.012	[0.933, 0.981]
		Moirai 2.0-Small	100	0.926	0.011	[0.904, 0.947]
		TiRex	100	0.934	0.011	[0.912, 0.956]
		Chronos-2	100	0.932	0.011	[0.911, 0.953]
	14	Chronos-Bolt-Tiny	100	0.956	0.012	[0.933, 0.978]
		Moirai 2.0-Small	100	0.928	0.011	[0.906, 0.948]
		TiRex	100	0.937	0.011	[0.915, 0.957]
		Chronos-2	100	0.934	0.011	[0.913, 0.954]
	28	Chronos-Bolt-Tiny	100	0.956	0.011	[0.934, 0.976]
		Moirai 2.0-Small	100	0.931	0.010	[0.910, 0.951]
		TiRex	100	0.942	0.010	[0.921, 0.961]
		Chronos-2	100	0.938	0.010	[0.917, 0.957]
Rohlik	7	Chronos-Bolt-Tiny	98	0.322	0.011	[0.301, 0.344]
		Moirai 2.0-Small	98	0.312	0.005	[0.302, 0.321]
		TiRex	98	0.314	0.011	[0.293, 0.336]
		Chronos-2	98	0.314	0.011	[0.293, 0.337]
	14	Chronos-Bolt-Tiny	98	0.334	0.012	[0.312, 0.358]
		Moirai 2.0-Small	98	0.327	0.011	[0.306, 0.349]
		TiRex	98	0.326	0.012	[0.305, 0.350]
		Chronos-2	98	0.327	0.012	[0.305, 0.351]
	28	Chronos-Bolt-Tiny	98	0.353	0.014	[0.327, 0.380]
		Moirai 2.0-Small	98	0.348	0.013	[0.323, 0.374]
		TiRex	98	0.345	0.013	[0.321, 0.372]
		Chronos-2	98	0.349	0.014	[0.323, 0.378]

Table D.18: Heuristic-weight diagnostic variants of the model-level log-ratio reanalysis ($k = 498$ Source A primary pool; legacy Source B descriptive).

Variant	k	$\hat{\ell}$	95% CI	p	Q
Best baseline Source A	498	−0.225	[−2.043, 1.592]	0.360	37,176
Average baseline	498	−0.286	[−2.525, 1.954]	0.352	39,209
Excl-M5	438	−0.226	[−2.075, 1.624]	0.365	22,578
WAPE-only	166	−0.157	[−1.720, 1.406]	0.423	9,839
SQL-only	166	−0.397	[−2.751, 1.956]	0.278	9,125
MASE-only	166	−0.124	[−1.575, 1.326]	0.473	7,411
Source B only	45	−0.197	[−0.814, 0.420]	0.154	7,101
Without fev-bench	84	−0.379	[−0.847, 0.089]	0.062	1,167
Without GIFT-Eval	414	−0.091	[−0.620, 0.437]	0.272	33,167
Equal-weight	498	−0.215	[−2.054, 1.623]	0.377	33

Appendix D.1. Summary Table

Appendix D.2. Interpretation

The sign is stable across sensitivity variants, but conservative suite-level containment intervals cross zero. This remains the key methodology repair: the benchmark rows provide many task-level comparisons, but they do not provide many independent source clusters. The Source B matched-panel rerun is useful local paired evidence, but it is not pooled into these Source A diagnostic variants.

Task-clustered CR2 sensitivity fits are more optimistic because they treat the 24 benchmark tasks as clusters. These are retained as software diagnostics only; they do not repair the missing paired sampling covariance.

Appendix D.3. Bucket-Level Variant ($k = 43$)

For comparison, a coarser bucket-level aggregation (mean FM vs mean ML_TREE per dataset–horizon–metric cell) yields $k = 43$ and $\hat{\ell} = -0.180$ (95% CI $[-0.313, -0.047]$, $p = 0.015$). This variant is retained only as a legacy sensitivity check; it averages away within-task model heterogeneity and is not the primary estimand.

Appendix E. Extraction Table (Abbreviated)

The extraction inputs (`analysis/extraction_schema.csv` and `benchmark/results/local_fm_sweep.csv`) record one row per (paper × model × dataset × metric) observation. The frozen extraction schema contains 802 rows; the local Source B sweep contributes 72 additional experiment rows. After canonicalising datasets, model families, and numeric metrics, the R pipeline retains 774 analysis-ready rows and forms $k = 498$ primary Source A paired comparisons plus 45 Source B descriptive contrasts. The complete CSVs are available in the supplementary materials. Table E.19 shows the key columns; columns omitted for space include `dataset_variant`, `n_series`, `series_length_median`, `frequency`, `fine_tuned`, `runtime_reported`, `gpu_hours`, `hardware`, and `notes`. The R pipeline additionally derives `study_id`, `suite_id`, and `task_id`; the resulting study characteristics table is written to `analysis/study_characteristics.csv`. That file summarises analysis-contributing sources and benchmark suites after canonicalisation; it is not a one-row-per-PRISMA-record catalogue of all 32 included external source records.

The official fev-bench prediction-level repair is stored separately from the frozen extraction table because it reruns benchmark windows rather than extracting published leaderboard rows. Its provenance is recorded in `analysis/figures/` through the official run manifest, model coverage, paired WAPE units, task contrasts, bootstrap draws, and covariance matrices. The generated file families follow the `fev_<model>_paired_wape_*.csv` pattern for Chronos-Bolt, Chronos-2, TiRex, and TimesFM 2.5.

Appendix E.1. Column Definitions

Table E.19: Key columns in the extraction table.

Column	Description
`paper_id`	Unique row/result identifier, not a dependency cluster
`study_id`	Derived publication or experiment-source identifier
`suite_id`	Derived source-suite identifier (fev-bench, GIFT-Eval, Source B, other)
`task_id`	Derived benchmark task/dataset/frequency/horizon cell identifier
`authors`	First author et al.
`year`	Publication year
`dataset`	Target dataset (M5, Favorita, Rohlik, GIFT-Eval, fev-bench, etc.)
`model_name`	Model as named by the paper
`model_family`	One of: foundation, deep_learning, ml_tree, statistical, ensemble
`zero_shot`	Yes / No — whether the model was applied without task-specific training
`horizon`	Forecast horizon (days) or "mixed" if aggregated
`eval_method`	rolling / fixed_origin / competition
`metric_name`	Reported metric (WAPE, WRMSSE, MASE, MAE, sMAPE, WQL, CRPS, win_rate_*)
`metric_value`	Reported value (numeric or qualitative rank)
`has_covariates`	Whether the model used exogenous covariates

Appendix E.2. Row Counts by Category

Appendix E.3. Source A vs Source B

- **Source A** (literature extraction): categories A–F (757 schema rows). The structured-search flow counts 32 included external study/source records; the larger row count comes from model–dataset–metric rows and per-task suite re-extraction, not from 757 independent studies. These rows report accuracy numbers as published by the original authors or benchmark repositories.
- **Source B** (gap-filling experiments): OWN + LOCAL rows. The current model-level analysis uses the 72-row `local_fm_sweep.csv` export from our own experiments on consumer hardware (MacBook M-series MPS) and Azure ML (E4DS_V4), using the shared protocol of §5.1; legacy OWN rows are retained in the extraction schema for provenance.

Table E.20: Extraction table row counts by category.

Category	Description	Rows
A	Foundation model papers	58
B	Benchmark papers and per-task re-extraction (incl. fev-bench, GIFT-Eval)	657
C	M5 competition and retail-specific	31
D	Cross-domain papers	3
F	Legacy fev-bench summary entries	8
OWN	Legacy Source B baseline rows in extraction schema	45
LOCAL	Local Source B sweep file (5 FMs + 3 baselines, MacBook/Azure)	72
Extraction schema		**802**
Local sweep file		**72**

The extraction table is frozen at the version used for the reanalysis of §6. Any post-submission additions would require re-running the generated contrast files and descriptive summaries of §6.

Appendix F. Threats to Validity

The analysis-contributing source assessment is generated as `analysis/risk_of_bias_assessment.csv`. It separates baseline quality, protocol matching, leakage risk, selective reporting, raw prediction availability, code availability, and independence notes for each `study_id`. Table F.21 summarises the resulting overall categories; Source B rows are not counted as included studies. This is a provenance and risk audit for the distinct sources that enter the analysis-ready extraction and benchmark reanalysis, not a full duplicate-screened risk-of-bias assessment for every manually screened candidate record or every one of the 32 Source A included studies.

Table F.21: Risk/provenance assessment summary for analysis-contributing sources. This is not a duplicate-screened risk-of-bias assessment for all PRISMA-included studies.

Suite	Overall risk category	Sources
GIFT-Eval	moderate	1
SourceB	high legacy / moderate repair	1
fev-bench	moderate	6
other	high/unclear	3

Five threats specific to the Source B foundation model runs (§5.4), in decreasing order of magnitude:

1. **Extraction selection bias.** Papers that publish on M5 / Favorita / Rohlik are not a random sample of the FM literature — they are the subset that chose a retail task

for their own evaluation. Papers that chose ETT / Weather / ECL (most of the LSF literature) are absent. The generated extraction and risk-of-bias artifacts surface this selection mechanism through source, suite, and provenance fields.

2. **Source A vs Source B protocol drift.** Source A numbers are extracted as-reported; Source B numbers are run on our §5.1.3 protocol. When the two diverge on the same (model, dataset, horizon) cell, the divergence is a data point about protocol sensitivity, not about model quality. The overlap between Source A and Source B is treated descriptively in §5.4.5, not as confirmatory paired evidence.

3. **Local run covers models up to 200 M params.** Moirai 2.0-Large (305 M) cannot be run locally (memory constraints, respectively) and is represented only by Source A. The five local models (Chronos-Bolt-Tiny 9 M, Moirai 2.0-Small 11 M, TiRex 35 M, Chronos-2 120 M, TimesFM 2.5 200 M) span over an order of magnitude in scale.

4. **Covariate asymmetry.** Most local FM families in Table 8 are univariate or are used in univariate zero-shot mode and cannot consume the §5.1.1 covariate sets. The LightGBM baselines in §5.3 use the full covariate sets. This biases the comparison toward LightGBM on covariate-driven datasets (M5: SNAP, prices; Favorita: oil, promotions) and is not correctable inside a zero-shot protocol. The moderator discussion in §6.8 treats covariate-aware vs univariate comparisons as hypothesis-generating.

5. **Top-30k Favorita cap propagates to the local run.** Same cap, same selection rule, same velocity bias as §5.1.1. Source A rows for Favorita include both fev-bench's top-54k subset and the full-Favorita subset where available; we flag `dataset_variant` per row so the §6 pool can restrict to comparable subsets.

Appendix G. Detailed Baseline Reliability Analysis

This appendix keeps the per-dataset baseline tables used by §5.3. The repaired analysis treats the direct-vs-recursive pattern as a fixed-budget diagnostic, not as a causal intermittency result. The stronger claims from earlier drafts are replaced by three audit conclusions: (i) the legacy fixed-budget protocol ranking differs across datasets, (ii) the matched-panel aggregate-WAPE repair makes M5 and Rohlik more conservative than the legacy per-series view and gives a mixed Favorita protocol ranking, and (iii) the legacy Favorita `direct_scaled` run should not be used as inferential evidence.

Appendix G.1. M5 Results

Table G.22 reports the rolling-origin M5 diagnostics. WRMSSE ranks direct LightGBM ahead of recursive LightGBM and Seasonal Naive within this protocol. These rows are not compared with the official M5 competition winner because the competition used a fixed 28-day task and a different evaluation setup.

On M5, direct LightGBM is better than recursive LightGBM on both WAPE and WRMSSE in the fixed-budget sweep. Seasonal Naive has lower per-series WAPE than direct LightGBM at every horizon, despite worse WRMSSE, which is why M5 WAPE is treated as a denominator-pathology warning rather than as the sole ranking metric.

Table G.22: Seasonal Naive vs LightGBM recursive vs LightGBM direct on the M5 tail-eval window. Bold = best per horizon.

Model	h	MAE	sMAPE	WAPE	WRMSSE	Runtime (s)	Cost (USD)
Seasonal Naive	7	1.1590	75.44	1.3072	0.7758	236.9	0.025
Seasonal Naive	14	1.1783	76.06	1.3216	0.7718	176.9	0.019
Seasonal Naive	28	1.1897	76.69	1.3285	0.7661	139.6	0.015
LightGBM rec	7	1.0110	144.44	1.6246	0.6336	300.3	0.032
LightGBM rec	14	1.0391	144.55	1.7292	0.6639	301.1	0.032
LightGBM rec	28	1.0743	144.48	1.9357	0.6976	284.3	0.030
LightGBM dir	7	**0.9682**	145.81	**1.3512**	**0.5598**	1114.4	0.118
LightGBM dir	14	**0.9896**	146.20	**1.4100**	**0.5821**	2003.2	0.212
LightGBM dir	28	**1.0117**	146.95	**1.5133**	**0.5992**	3728.8	0.394

Appendix G.2. Rohlik Results

Table G.23: Seasonal Naive vs LightGBM recursive vs LightGBM direct on the Rohlik v2 tail-eval window. Bold = best per horizon among LightGBM variants.

Model	h	MAE	sMAPE	WAPE	Runtime (s)	Cost (USD)
Seasonal Naive	7	40.649	36.46	0.4015	8.5	0.0009
Seasonal Naive	14	42.236	37.20	0.4031	6.9	0.0007
Seasonal Naive	28	43.543	38.21	0.4116	5.8	0.0006
LightGBM rec	7	**19.335**	92.90	**0.3654**	29.6	0.0031
LightGBM rec	14	20.972	94.59	**0.3953**	30.6	0.0032
LightGBM rec	28	22.815	97.18	**0.4321**	29.1	0.0031
LightGBM dir	7	19.549	93.34	0.3816	141.1	0.0149
LightGBM dir	14	**20.210**	94.17	0.4072	265.6	0.0280
LightGBM dir	28	**21.087**	95.01	0.4442	524.3	0.0553

Rohlik reverses the M5 fixed-budget WAPE ranking: recursive LightGBM is lower than direct at all three horizons, while direct is several times more expensive. This is a protocol-and-dataset interaction hypothesis, not a standalone intermittency proof.

Appendix G.3. Favorita Results And `direct_scaled` Flag

In the fixed-budget Favorita sweep, recursive LightGBM is cheaper and lower on WAPE than direct LightGBM. The follow-up `direct_scaled_favorita.csv` run is not interpreted as a resolved robustness result. It records WAPE 0.4878, 0.5060, and 0.5264 at $h = 7, 14, 28$, which is lower than recursive at all horizons. However, the $h = 7$ scaled run still has 300 trees per head, the same nominal count as the base direct run, while WAPE changes from 0.5513 to 0.4878. This is flagged in the generated `source_b_direct_scaled_audit.csv` file as the $h = 7$ same-tree-count inconsistency. Until the exact protocol difference is recovered, the scaled run is a rerun target rather than evidence for a new ranking.

Table G.24: Seasonal Naive vs LightGBM recursive vs LightGBM direct on the Favorita top-30k tail-eval window. Bold = best per horizon among LightGBM variants.

Model	h	MAE	sMAPE	WAPE	Runtime (s)	Cost (USD)
Seasonal Naive	7	5.248	60.76	0.6363	65	0.0068
Seasonal Naive	14	5.400	61.49	0.6473	49	0.0052
Seasonal Naive	28	5.644	62.49	0.6643	39	0.0041
LightGBM rec	7	**2.507**	90.92	**0.5295**	236	0.0249
LightGBM rec	14	**2.574**	91.32	**0.5311**	230	0.0242
LightGBM rec	28	2.692	92.13	**0.5377**	230	0.0243
LightGBM dir	7	2.533	91.38	0.5513	1,015	0.1071
LightGBM dir	14	2.619	92.17	0.5711	1,907	0.2012
LightGBM dir	28	**2.688**	92.75	0.5892	3,588	0.3787

Table G.25: Legacy full-workload summary of the fixed-budget direct-vs-recursive LightGBM gap using per-series WAPE means (positive = direct lower, negative = recursive lower).

Dataset	Zero-day frac.	$h = 7$	$h = 14$	$h = 28$	Interpretation
M5	∼70 %	+16.8 %	+18.5 %	+21.8 %	Direct lower in this protocol
Rohlik	∼1.2 %	−4.4 %	−3.0 %	−2.8 %	Recursive lower and cheaper
Favorita	∼15–25 %	−4.1 %	−7.5 %	−9.6 %	Recursive lower in fixed-budget run

Appendix G.4. Legacy Fixed-Budget Protocol Summary

Appendix G.5. Matched-Panel Protocol Check

The matched-panel repair recomputes the M5, Rohlik, and Favorita LightGBM protocol contrast on the same 100 top-volume series used for the FM-vs-baseline paired panel. Using aggregate WAPE, direct is slightly worse than recursive on M5 at $h = 7$ and $h = 14$, slightly better at $h = 28$, and worse than recursive on Rohlik at all horizons. On Favorita, recursive is lower at $h = 7$ and $h = 14$, while direct is lower at $h = 28$. The horizon-scaled direct run equals the unscaled direct run at $h = 7$ on all three datasets, as expected when both use 300 trees. This resolves the same-tree-count anomaly in the repaired panel.

The combined evidence supports a cautious hypothesis: the best LightGBM protocol can change with the dataset, metric aggregation, sampled workload, and training budget. It does not support a hard intermittency threshold, nor a general rule that direct or recursive is always preferable.

Appendix G.6. Cost Consistency Audit

The Azure cost tables are retained for audit, but this paper does not use them for universal cost conclusions. The detail tables and summary table do not yet reconcile to one ledger.

The cost plots in §6.7 therefore remain cost-error scatterplots. The matched-panel repair writes a local run ledger for the completed M5, Rohlik, and Favorita reruns. The older Azure summary/detail tables still need a single reconciled ledger before cost claims are promoted beyond descriptive accounting.

Table G.26: Baseline cost consistency audit. Differences are too large to treat as rounding; a single run ledger is required before cost claims are promoted from descriptive to inferential.

Scope	Detail tables (USD)	Summary table (USD)	Difference	Status
M5 total	0.877	0.810	−0.067	mismatch requires single ledger
Rohlik total	0.110	0.120	+0.010	mismatch requires single ledger
Favorita total	0.777	1.119	+0.343	mismatch requires single ledger
Favorita direct	0.687	1.030	+0.343	mismatch requires single ledger
M5 seasonal naive	0.059	0.146	+0.087	mismatch requires single ledger
M5 direct	0.724	0.571	−0.153	mismatch requires single ledger

Appendix G.7. Reproducibility Detail

Every baseline result should be tied to a git commit SHA, an Azure ML pipeline run name, a registered data asset version, and a versioned environment. The matched-panel runs store shared series identifiers, forecast origins, actuals, per-model predictions, and paired bootstrap contrasts for completed M5, Rohlik, and Favorita cells. The same requirement applies to future GPU/larger-workload reruns and covariate-aware FM variants before Source B can enter a broader formal paired analysis.

Declaration of competing interest

The author declares no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Maciej Rubczynski: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

Data availability

A historical technical-report snapshot is archived on Zenodo at `https://doi.org/10.5281/zenodo.21338004`. The live repository for the current analysis is `https://github.`

com/MaciejR/large-scale-demand-forecasting-benchmark; the release package corresponding to this manuscript is prepared as a versioned Zenodo-ready archive under release/ rather than overwriting the historical snapshot, and is published as GitHub release v1.12 at https://github.com/MaciejR/large-scale-demand-forecasting-benchmark/releases/tag/v1.12. The package records the source commit in COMMIT.txt and file hashes in CHECKSUMS.txt. The repository includes derived extraction tables, analysis code, generated tables/figures, study-characteristics data, audit outputs, and replication scripts, but not raw competition data or untracked fev-bench prediction parquet trees. The M5 dataset is publicly available via Kaggle; the Favorita dataset is publicly available via Kaggle; the Rohlik v2 dataset is available at https://github.com/Rohlik/rohlik-orders-forecasting-challenge.

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